

reproduce__LeverageSHAP__custom__vs__LeverageSHAPWo2c

August 6, 2026

```
[1]: # %%  
# ruff: noqa: T201  
"""Benchmark evaluating the isolated effect of the 2c threshold in LeverageSHAP.  
  
The comparison includes:  
1. KernelSHAP (Standard Control)  
2. LeverageSHAP (Without 2c Limit) - Leverage score weights, but uniform budget  
   ↪ allocation  
3. LeverageSHAP (Full Custom) - Optimal 2c-based Bernoulli thresholding  
"""  
  
from __future__ import annotations  
  
import math  
import random as _py_random  
import warnings  
from typing import Any  
  
import matplotlib.pyplot as plt  
import numpy as np  
import xgboost as xgb  
from scipy.stats import wilcoxon  
from sklearn.datasets import fetch_california_housing  
from sklearn.model_selection import train_test_split  
  
from shapiq import KernelSHAP  
from shapiq.approximator.regression.leverageshap import LeverageSHAP  
from shapiq.approximator.sampling import CoalitionSampler  
from shapiq.interaction_values import InteractionValues  
  
warnings.filterwarnings("ignore")
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[2]: # %%  
# LeverageSHAP Variant without the 2c Threshold Limit  
class LeverageSHAPWithout2c(LeverageSHAP):  
    """LeverageSHAP variant that bypasses the 2c oversampling and threshold  
    ↪ boundaries.
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    This class overrides the sampling method to demonstrate the impact of
    ↪omitting
    the optimal boundary thresholding from the algorithm.
    """

    def _sample(self, budget: int) -> tuple[np.ndarray, np.ndarray]:
        if budget < 2:
            msg = "Budget must be at least 2 to evaluate baseline and grand
            ↪coalition."
            raise ValueError(msg)

        n = self.n
        m = min(budget, 2*n)

        z_empty = np.zeros(n, dtype=bool)
        z_grand = np.ones(n, dtype=bool)

        # -----
        # - OMITTED: Finding the optimal oversampling parameter `c` via binary
        ↪search
        # - c = self._find_c(n, m)
        # -----
        # + ADDED INSTEAD:
        # + Without the parameter `c`, a flat allocation is applied to
        ↪distribute
        # + the total pair budget (m - 2) // 2 evenly across all possible size
        ↪layers s.
        # -----
        n_pairs_target = (m - 2) // 2
        sizes_to_sample = list(range(1, n // 2 + 1))
        num_sizes = len(sizes_to_sample)

        pairs_per_size = n_pairs_target // num_sizes if num_sizes > 0 else 0
        remainder = n_pairs_target % num_sizes if num_sizes > 0 else 0

        z_list: list[np.ndarray] = []
        sizes_list: list[int] = []

        py_seed = int(self._rng.integers(0, 2**32))
        py_rng = _py_random.Random(py_seed)

        for idx_size, s in enumerate(sizes_to_sample):
            is_middle = (n % 2 == 0) and (s == n // 2)
            full_count = math.comb(n, s)
            pool_size = math.comb(n - 1, s - 1) if is_middle else full_count

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# Determine target pair budget for this size layer
m_s_target = pairs_per_size + (1 if idx_size < remainder else 0)
m_s = min(m_s_target, pool_size)

# -----
# - OMITTED: Bernoulli sampling using 2c-based probability and
↳ thresholding
# - Z_pairs, sizes = self._bernoulli_sample(n, c)
# - (Which caps smaller sizes at prob = 1.0 and samples larger
↳ sizes with prob = 2c / full_count)
# -----
# + ADDED INSTEAD:
# + We bypass the binomial probabilistic assignment and sample
↳ exactly `m_s`
# + unique coalitions directly without threshold limits.
# -----
indices = self._sample_without_replacement(pool_size, m_s, py_rng)

for idx in indices:
    if is_middle:
        z_partial = self._combo(n - 1, s - 1, idx)
        z = np.zeros(n, dtype=bool)
        z[: n - 1] = z_partial
        z[n - 1] = True
    else:
        z = self._combo(n, s, idx)
    z_bar = ~z
    z_list.append(z)
    z_list.append(z_bar)
    sizes_list.append(s)
    sizes_list.append(n - s)

if len(z_list) > 0:
    Z_pairs = np.array(z_list)
    sizes = np.array(sizes_list)
    weights_pairs = np.empty(Z_pairs.shape[0], dtype=float)
    fact_n = math.factorial(n)

    for i, s in enumerate(sizes):
        w_s = (math.factorial(s - 1) * math.factorial(n - s - 1)) /
↳ fact_n

#
↳ -----
# - OMITTED: Importance sampling reweighting using 2c and
↳ leverage score

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# - l_z = 1.0 / math.comb(n, s)
# - p = min(1.0, 2.0 * c * l_z)
# - weights_pairs[i] = w_s / p
#_

↪ -----

# + ADDED INSTEAD:
# + Without `c`, we use the actual sampling probability of_
↪ drawing
# + a specific coalition in this size layer: p = m_s /_
↪ pool_size.
# + This maintains the weight-cancellation property of the_
↪ leverage scores
# + but does not use the optimal planning constant `c`.
#_

↪ -----

s_paired = min(s, n - s)
is_middle_s = (n % 2 == 0) and (s_paired == n // 2)
pool_s = math.comb(n - 1, s_paired - 1) if is_middle_s else_
↪ math.comb(n, s_paired)

idx_in_loop = sizes_to_sample.index(s_paired)
m_s_drawn = pairs_per_size + (1 if idx_in_loop < remainder else_
↪ 0)

m_s_drawn = min(m_s_drawn, pool_s)

p_s_actual = m_s_drawn / pool_s if pool_s > 0 else 1.0
if p_s_actual <= 0:
    p_s_actual = 1.0

weights_pairs[i] = w_s / p_s_actual

Z = np.vstack([z_empty[None, :], z_grand[None, :], Z_pairs])
weights = np.concatenate([[0.0, 0.0], weights_pairs])
else:
    Z = np.vstack([z_empty[None, :], z_grand[None, :]])
    weights = np.array([0.0, 0.0])

return Z, weights

class LeverageSHAPWoBernoulli(LeverageSHAP):
    """LeverageSHAP using the standard CoalitionSampler instead of Algorithm 2.
    ↪ """

    def __init__(

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        self, n: int, *, pairing_trick: bool = True, random_state: int | None = None
→None, **kwargs: Any
    ) -> None:
        """Initialize the class override for benchmarking."""
        self._explicit_seed = random_state
        super().__init__(n=n, pairing_trick=pairing_trick, _
→random_state=random_state, **kwargs)

    def _sample(self, budget: int) -> tuple[np.ndarray, np.ndarray]:
        sampler = CoalitionSampler(
            n_players=self.n,
            sampling_weights=np.ones(self.n + 1),
            pairing_trick=True,
            random_state=self._explicit_seed,
        )

        sampler.sample(budget)
        Z = sampler.coalitions_matrix
        is_weights = sampler.sampling_adjustment_weights

        weights = np.zeros_like(is_weights)
        fact_n = math.factorial(self.n)
        sizes = Z.sum(axis=1)

        for i, s in enumerate(sizes):
            if 0 < s < self.n:
                w_s = (math.factorial(s - 1) * math.factorial(self.n - s - 1)) /
→ fact_n
                weights[i] = is_weights[i] * w_s

        return Z, weights

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[3]: # %%
# Metrics & Extraction Helpers
def norm_l2(exact: np.ndarray, approx: np.ndarray) -> float:
    """Normalized error: ||exact - approx|| / ||exact|| ."""
    denom = np.linalg.norm(exact)
    return 0.0 if denom < 1e-12 else float(np.linalg.norm(exact - approx) / _
→denom)

def extract_sv(iv: InteractionValues, n: int) -> np.ndarray:
    """Extract Shapley vector of length n from an IV mapping."""
    return np.array([iv[(i,)] for i in range(n)])

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[4]: # %%
# Setup California Dataset (n=8)
X, y = fetch_california_housing(return_X_y=True)
n_players = 8

X_train, X_test, y_train, _ = train_test_split(X[:, :n_players], y, test_size=0.
↪2, random_state=42)

model = xgb.XGBRegressor(n_estimators=100, max_depth=4, random_state=42,
↪verbosity=0)
model.fit(X_train, y_train)

bg_mean = X_train.mean(axis=0)
x_instance = X_test[0]

def game(Z: np.ndarray) -> np.ndarray:
    """Mean-substitution game."""
    X_masked = np.where(Z, x_instance[np.newaxis, :], bg_mean[np.newaxis, :])
    return model.predict(X_masked)
```

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[5]: # %%
# Run the Benchmark
n_runs = 50
budgets = np.linspace(10, 150, 15, dtype=int).tolist()

errs_ks = np.zeros((n_runs, len(budgets)))
errs_lev_no_2c = np.zeros((n_runs, len(budgets)))
errs_lev_full = np.zeros((n_runs, len(budgets)))
errs_lev_wo_bernoulli = np.zeros((n_runs, len(budgets)))

# Ground Truth Exact values
ks_exact = KernelSHAP(n=n_players, random_state=0)
iv_exact = ks_exact.approximate(2*n_players, game)
exact_sv = extract_sv(iv_exact, n_players)

print("Starting benchmark calculation...")
for i, budget in enumerate(budgets):
    for seed in range(n_runs):
        # 1. KernelSHAP (Control)
        ks = KernelSHAP(n=n_players, random_state=seed)
        errs_ks[seed, i] = norm_l2(exact_sv, extract_sv(ks.approximate(budget,
↪game), n_players))

        # 2. LeverageSHAP without the 2c limit
        lev_no_2c = LeverageSHAPWithout2c(n=n_players, random_state=seed)
        errs_lev_no_2c[seed, i] = norm_l2(
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        exact_sv, extract_sv(lev_no_2c.approximate(budget, game), n_players)
    )

    # 3. Full Custom LeverageSHAP (with 2c)
    lev_full = LeverageSHAP(n=n_players, random_state=seed)
    errs_lev_full[seed, i] = norm_l2(
        exact_sv, extract_sv(lev_full.approximate(budget, game), n_players)
    )

    # 4. LeverageSHAPWoBernoulli
    lev_wo_bernoulli = LeverageSHAPWoBernoulli(n=n_players,
    ↪random_state=seed)
    errs_lev_wo_bernoulli[seed, i] = norm_l2(
        exact_sv, extract_sv(lev_wo_bernoulli.approximate(budget, game),
    ↪n_players)
    )

```

Starting benchmark calculation...

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[6]: # %%
# Plotting Helper
def plot_results(plot_kernel: bool = False) -> None:
    plt.figure(figsize=(10, 6))

    if plot_kernel:
        plt.semilogy(
            budgets,
            np.median(errs_ks, axis=0),
            "o-.",
            label="KernelSHAP (Control Baseline)",
            color="blue",
            linewidth=2,
        )

    plt.semilogy(
        budgets,
        np.median(errs_lev_no_2c, axis=0),
        "x--",
        label="LeverageSHAP (Without 2c Limit)",
        color="orange",
        linewidth=2,
    )

    plt.semilogy(
        budgets,
        np.median(errs_lev_full, axis=0),
        "s-",

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        label="LeverageSHAP (Full Custom)",
        color="green",
        linewidth=2.5,
    )

plt.semilogy(
    budgets,
    np.median(errs_lev_wo_bernoulli, axis=0),
    "d--",
    label="LeverageSHAP (Wo Bernoulli)",
    color="red",
    linewidth=2,
)

# Fill Quantiles
if plot_kernel:
    plt.fill_between(
        budgets,
        np.percentile(errs_ks, 25, axis=0),
        np.percentile(errs_ks, 75, axis=0),
        alpha=0.1,
        color="blue",
    )

plt.fill_between(
    budgets,
    np.percentile(errs_lev_no_2c, 25, axis=0),
    np.percentile(errs_lev_no_2c, 75, axis=0),
    alpha=0.1,
    color="orange",
)

plt.fill_between(
    budgets,
    np.percentile(errs_lev_full, 25, axis=0),
    np.percentile(errs_lev_full, 75, axis=0),
    alpha=0.15,
    color="green",
)

plt.fill_between(
    budgets,
    np.percentile(errs_lev_wo_bernoulli, 25, axis=0),
    np.percentile(errs_lev_wo_bernoulli, 75, axis=0),
    alpha=0.1,
    color="red",
)

plt.xlabel("Sample Size (Budget m)")

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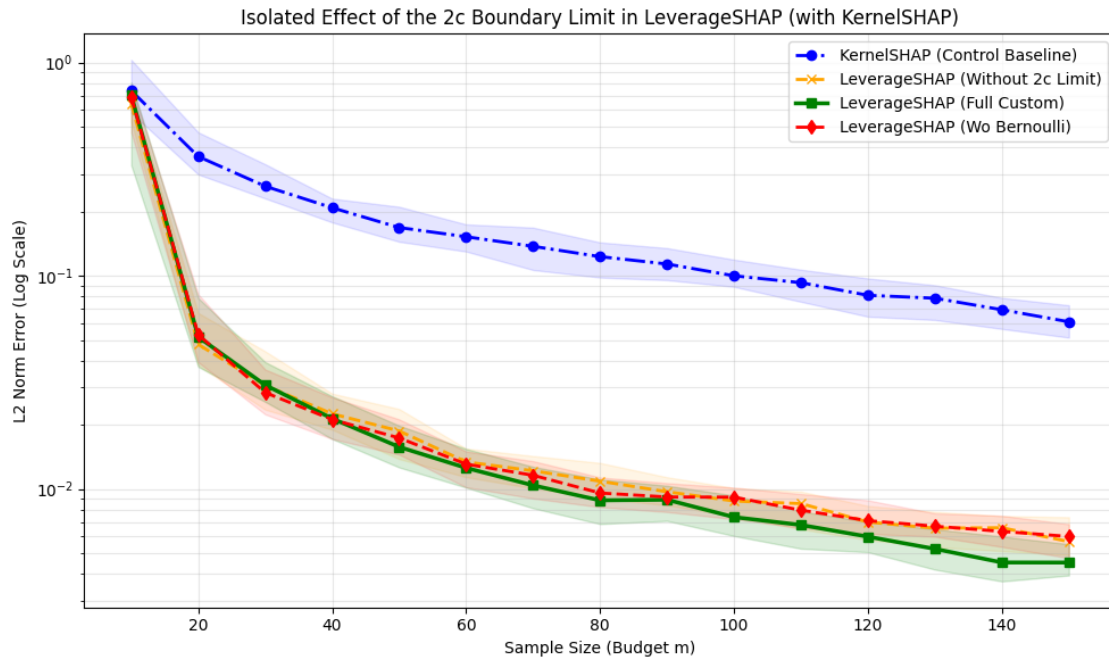


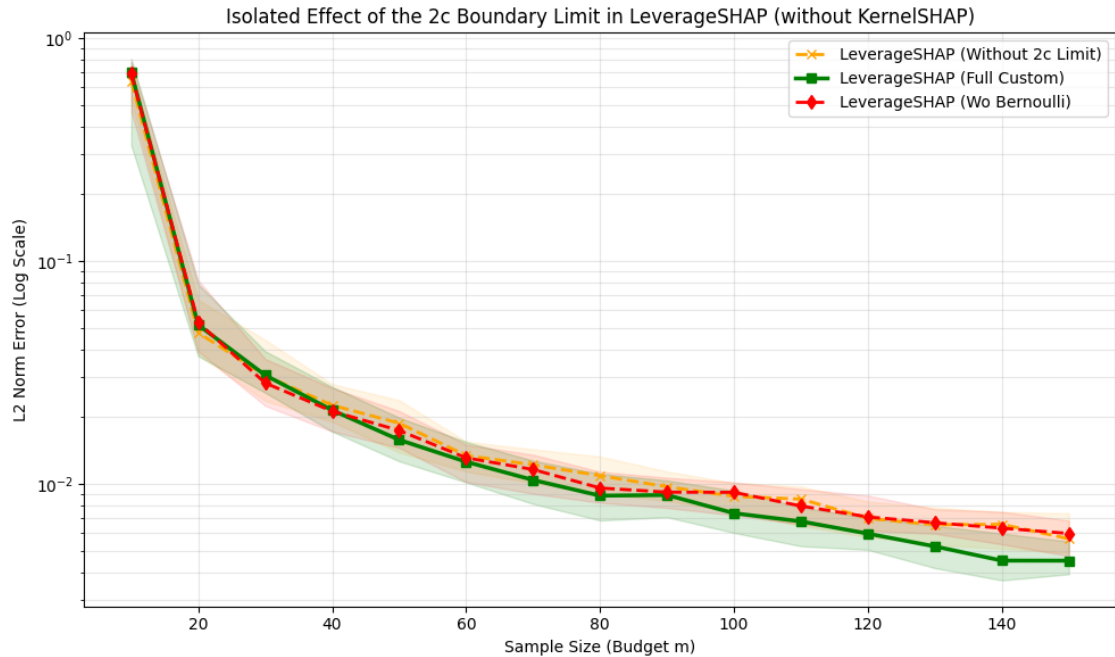
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plt.ylabel("L2 Norm Error (Log Scale)")
plt.title(
    "Isolated Effect of the 2c Boundary Limit in LeverageSHAP"
    + (" (with KernelSHAP)" if plot_kernel else " (without KernelSHAP)")
)
plt.legend()
plt.grid(True, which="both", alpha=0.3)
plt.tight_layout()
plt.show()

# Run both variants
plot_results(plot_kernel=True)
plot_results(plot_kernel=False)

```





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[7]: import pandas as pd

# Build per-instance table (each run x budget pair)
n_runs_, n_budgets_ = errs_lev_no_2c.shape

table = pd.DataFrame(
    {
        "run": np.repeat(np.arange(n_runs_), n_budgets_),
        "budget": np.tile(budgets, n_runs_),
        "err_ks": errs_ks.ravel(),
        "err_lev_no_2c": errs_lev_no_2c.ravel(),
        "err_lev_full": errs_lev_full.ravel(),
        "err_lev_wo_bernoulli": errs_lev_wo_bernoulli.ravel(),
    }
)

# Wins (lower error is better)
table["lev_full_wins_vs_no_2c"] = table["err_lev_full"] < table["err_lev_no_2c"]
table["lev_full_wins_vs_wo_bernoulli"] = table["err_lev_full"] <_
    ↪ table["err_lev_wo_bernoulli"]
table["no_2c_wins_vs_wo_bernoulli"] = table["err_lev_no_2c"] <_
    ↪ table["err_lev_wo_bernoulli"]

# Win margins (positive => left model is better)
table["margin_full_vs_no2c"] = table["err_lev_no_2c"] - table["err_lev_full"]
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table["margin_full_vs_wo_bern"] = table["err_lev_wo_bernoulli"] -
    ↪table["err_lev_full"]
table["margin_no2c_vs_wo_bern"] = table["err_lev_wo_bernoulli"] -
    ↪table["err_lev_no_2c"]

print("=== Per-instance table (all run x budget entries) ===")
print(table.to_string(index=False))

# Budget-wise win rates AND paired significance
# The 750 rows are NOT independent: the same 50 seeds recur across all 15
    ↪budgets and
# are shared by every method. Treating them as 750 independent Bernoulli trials
    ↪would
# overstate precision. We therefore report a paired Wilcoxon signed-rank test
    ↪*per
# budget* on the per-seed error differences (the 50 seeds are the paired units
    ↪within a
# budget). "Sig." =  $p < 0.05$  AND lower median error for lev_full.
def paired_wilcoxon(err_a: np.ndarray, err_b: np.ndarray) -> tuple[float,
    ↪float]:
    """Return (median difference b - a, p-value) for a paired Wilcoxon test per
    ↪column-slice."""
    diff = err_b - err_a # positive => a (lev_full) is better
    if np.allclose(diff, 0.0):
        return 0.0, 1.0
    try:
        _, p = wilcoxon(diff)
    except ValueError:
        return float(np.median(diff)), 1.0
    return float(np.median(diff)), float(p)

sig_rows = []
for i, budget in enumerate(budgets):
    med_fn, p_fn = paired_wilcoxon(errs_lev_full[:, i], errs_lev_no_2c[:, i])
    med_fb, p_fb = paired_wilcoxon(errs_lev_full[:, i], errs_lev_wo_bernoulli[
    ↪, i])
    sig_rows.append(
        {
            "budget": budget,
            "p_full_vs_no2c": round(p_fn, 4),
            "full_better_no2c": bool(p_fn < 0.05 and med_fn > 0),
            "p_full_vs_wo_bern": round(p_fb, 4),
            "full_better_wo_bern": bool(p_fb < 0.05 and med_fb > 0),
        }
    )

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    )
sig_by_budget = pd.DataFrame(sig_rows)

# Budget-wise win summary (raw win counts, kept for context; see significance_
↳table above)
summary_by_budget = table.groupby("budget", as_index=False).agg(
    instances=("budget", "size"),
    lev_full_win_count=("lev_full_wins_vs_no_2c", "sum"),
    lev_full_vs_wo_bern_win_count=("lev_full_wins_vs_wo_bernoulli", "sum"),
    no_2c_win_count=("no_2c_wins_vs_wo_bernoulli", "sum"),
    median_err_lev_full=("err_lev_full", "median"),
    median_err_lev_no_2c=("err_lev_no_2c", "median"),
    median_err_lev_wo_bernoulli=("err_lev_wo_bernoulli", "median"),
)
summary_by_budget = summary_by_budget.merge(sig_by_budget, on="budget")

print("\n=== Paired Wilcoxon significance by budget (50 shared seeds per_
↳budget) ===")
print(sig_by_budget.to_string(index=False))

n_sig_full_no2c = int(sig_by_budget["full_better_no2c"].sum())
n_sig_full_wo_bern = int(sig_by_budget["full_better_wo_bern"].sum())
print(
    f"\nlev_full significantly better than lev_no_2c at {n_sig_full_no2c}/
↳{len(budgets)} budgets; "
    f"than lev_wo_bernoulli at {n_sig_full_wo_bern}/{len(budgets)} budgets (p <_
↳0.05, paired).")
)

# Overall paired test: pool the per-seed differences across budgets but respect_
↳pairing
# (each (seed, budget) difference is one paired observation).
overall_rows = []
for name, err_left, err_right in [
    ("lev_full vs lev_no_2c", errs_lev_full, errs_lev_no_2c),
    ("lev_full vs lev_wo_bernoulli", errs_lev_full, errs_lev_wo_bernoulli),
    ("lev_no_2c vs lev_wo_bernoulli", errs_lev_no_2c, errs_lev_wo_bernoulli),
]:
    diff = (err_right - err_left).ravel() # positive => left is better
    try:
        _, p = wilcoxon(diff)
    except ValueError:
        p = 1.0
    overall_rows.append(
        {
            "comparison": name,
            "median_margin": round(float(np.median(diff)), 5),

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        "p_value": f"{p:.2e}",
        "left_better": bool(p < 0.05 and np.median(diff) > 0),
    }
)
overall = pd.DataFrame(overall_rows)

print("\n=== Overall paired Wilcoxon summary (positive margin => left model_
↳better) ===")
print(overall.to_string(index=False))

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=== Per-instance table (all run x budget entries) ===
run budget      err_ks err_lev_no_2c err_lev_full err_lev_wo_bernoulli
lev_full_wins_vs_no_2c lev_full_wins_vs_wo_bernoulli
no_2c_wins_vs_wo_bernoulli margin_full_vs_no2c margin_full_vs_wo_bern
margin_no2c_vs_wo_bern
0      10 9.387193e-01      0.701070      0.847305      0.791678
False
-1.462345e-01      -0.055626      9.060822e-02
0      20 4.137244e-01      0.038156      0.813323      0.093654
False
-7.751674e-01      -0.719669      5.549824e-02
0      30 1.849863e-01      0.045829      0.039031      0.041928
True
6.798088e-03      0.002897      -3.900779e-03
0      40 1.722260e-01      0.022148      0.021287      0.016682
True
8.608494e-04      -0.004605      -5.465499e-03
0      50 1.362410e-01      0.019493      0.023753      0.015707
False
-4.259937e-03      -0.008046      -3.785949e-03
0      60 9.743068e-02      0.008141      0.020842      0.009998
False
-1.270089e-02      -0.010844      1.857035e-03
0      70 9.424941e-02      0.009165      0.014011      0.010255
False
-4.845806e-03      -0.003757      1.089061e-03
0      80 1.122590e-01      0.009964      0.015470      0.006349
False
-5.506548e-03      -0.009121      -3.614298e-03
0      90 1.354771e-01      0.007621      0.009386      0.004595
False
-1.765391e-03      -0.004791      -3.025658e-03
0      100 1.241587e-01      0.008361      0.008458      0.006183
False
-9.650064e-05      -0.002275      -2.178137e-03
0      110 7.346209e-02      0.009434      0.003906      0.005880
True
5.528781e-03      0.001974      -3.554561e-03

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0	120	4.727594e-02	0.006380	0.006676	0.005163
False			False	False	
-2.965313e-04			-0.001513	-1.216942e-03	
0	130	5.111932e-02	0.005375	0.005910	0.005721
False			False	True	
-5.350201e-04			-0.000189	3.458240e-04	
0	140	9.868267e-02	0.007005	0.007409	0.005464
False			False	False	
-4.039045e-04			-0.001945	-1.540687e-03	
0	150	1.183154e-01	0.003623	0.006452	0.004062
False			False	True	
-2.828774e-03			-0.002389	4.394453e-04	
1	10	5.375910e-01	0.067278	0.055016	0.712611
True			True	True	
1.226199e-02			0.657595	6.453329e-01	
1	20	5.724136e-01	0.066100	0.050580	0.045869
True			False	False	
1.552087e-02			-0.004711	-2.023182e-02	
1	30	4.131897e-01	0.025724	0.018017	0.018535
True			True	False	
7.706650e-03			0.000518	-7.188899e-03	
1	40	1.864106e-01	0.025776	0.028642	0.015505
False			False	False	
-2.866296e-03			-0.013137	-1.027116e-02	
1	50	1.456904e-01	0.019706	0.029742	0.017214
False			False	False	
-1.003551e-02			-0.012528	-2.492043e-03	
1	60	1.279519e-01	0.012112	0.012419	0.014952
False			True	True	
-3.067194e-04			0.002533	2.839606e-03	
1	70	1.166629e-01	0.009943	0.008037	0.008770
True			True	False	
1.905474e-03			0.000733	-1.172534e-03	
1	80	7.841726e-02	0.010822	0.007634	0.008175
True			True	False	
3.187862e-03			0.000541	-2.647133e-03	
1	90	6.871282e-02	0.011336	0.009732	0.006670
True			False	False	
1.604414e-03			-0.003061	-4.665879e-03	
1	100	8.966786e-02	0.010448	0.010296	0.008280
True			False	False	
1.525574e-04			-0.002016	-2.168788e-03	
1	110	9.009210e-02	0.009056	0.006537	0.009363
True			True	True	
2.519066e-03			0.002826	3.069455e-04	
1	120	6.598653e-02	0.005812	0.002700	0.006604
True			True	True	
3.111907e-03			0.003904	7.917517e-04	

1	130	5.529107e-02	0.003242	0.003400	0.006626
False			True	True	
-1.582725e-04			0.003226	3.384252e-03	
1	140	7.880565e-02	0.002201	0.003719	0.006305
False			True	True	
-1.517572e-03			0.002586	4.103483e-03	
1	150	5.842021e-02	0.004045	0.004424	0.006235
False			True	True	
-3.794914e-04			0.001811	2.190332e-03	
2	10	5.265511e-01	0.627242	0.736191	0.657394
False			False	True	
-1.089487e-01			-0.078796	3.015221e-02	
2	20	4.944379e-01	0.055431	0.043990	0.055431
True			True	True	
1.144049e-02			0.011441	1.335089e-07	
2	30	2.510270e-01	0.098008	0.038439	0.049166
True			True	False	
5.956823e-02			0.010727	-4.884138e-02	
2	40	3.189226e-01	0.026298	0.027368	0.040663
False			True	True	
-1.070172e-03			0.013295	1.436504e-02	
2	50	2.142400e-01	0.013215	0.008040	0.010120
True			True	False	
5.175209e-03			0.002080	-3.095137e-03	
2	60	1.537664e-01	0.009012	0.004566	0.018828
True			True	True	
4.445289e-03			0.014261	9.816138e-03	
2	70	1.567920e-01	0.009476	0.008649	0.019550
True			True	True	
8.261552e-04			0.010900	1.007401e-02	
2	80	1.186102e-01	0.004726	0.006655	0.008698
False			True	True	
-1.929292e-03			0.002043	3.972166e-03	
2	90	1.260823e-01	0.004840	0.006085	0.009219
False			True	True	
-1.244668e-03			0.003135	4.379203e-03	
2	100	9.018854e-02	0.002538	0.006385	0.009309
False			True	True	
-3.847308e-03			0.002924	6.771515e-03	
2	110	1.001026e-01	0.008745	0.005472	0.006645
True			True	False	
3.272985e-03			0.001173	-2.100282e-03	
2	120	9.311918e-02	0.008085	0.008472	0.006347
False			False	False	
-3.873451e-04			-0.002125	-1.738057e-03	
2	130	8.032533e-02	0.007700	0.005528	0.006010
True			True	False	
2.171345e-03			0.000482	-1.689788e-03	

2	140	4.700572e-02	0.008339	0.006681	0.005213
True			False		False
1.658673e-03			-0.001468		-3.126589e-03
2	150	5.327451e-02	0.005086	0.006637	0.005007
False			False		False
-1.551114e-03			-0.001631		-7.966925e-05
3	10	2.986944e-01	0.700937	0.709115	0.075292
False			False		False
-8.177329e-03			-0.633822		-6.256449e-01
3	20	3.902203e-01	0.657334	0.053795	0.070737
True			True		False
6.035395e-01			0.016942		-5.865971e-01
3	30	2.558112e-01	0.024265	0.035444	0.029456
False			False		True
-1.117944e-02			-0.005988		5.191246e-03
3	40	2.021357e-01	0.017701	0.020057	0.020297
False			True		True
-2.356373e-03			0.000240		2.596264e-03
3	50	2.182825e-01	0.014895	0.020605	0.011715
False			False		False
-5.709994e-03			-0.008891		-3.180888e-03
3	60	1.785874e-01	0.012352	0.009638	0.007480
True			False		False
2.714378e-03			-0.002157		-4.871870e-03
3	70	1.445238e-01	0.012932	0.006628	0.008575
True			True		False
6.304188e-03			0.001947		-4.357022e-03
3	80	1.325928e-01	0.011253	0.008601	0.010358
True			True		False
2.651761e-03			0.001757		-8.951169e-04
3	90	1.465727e-01	0.006882	0.006685	0.007765
True			True		True
1.969777e-04			0.001080		8.834937e-04
3	100	1.237035e-01	0.007650	0.007766	0.009473
False			True		True
-1.163168e-04			0.001706		1.822671e-03
3	110	1.125544e-01	0.005255	0.008528	0.007899
False			False		True
-3.272868e-03			-0.000629		2.644153e-03
3	120	6.596121e-02	0.006082	0.005550	0.007384
True			True		True
5.324740e-04			0.001835		1.302352e-03
3	130	6.720299e-02	0.005037	0.004111	0.006835
True			True		True
9.265017e-04			0.002724		1.797171e-03
3	140	7.450693e-02	0.006715	0.006014	0.005680
True			False		False
7.013261e-04			-0.000334		-1.035064e-03

3	150	6.522842e-02	0.009296	0.004543	0.003265
True			False		False
4.752930e-03			-0.001278		-6.030582e-03
4	10	6.128375e-01	0.029243	0.425558	0.788840
False			True		True
-3.963143e-01			0.363282		7.595965e-01
4	20	7.454540e-01	0.048157	0.011075	0.067764
True			True		True
3.708266e-02			0.056690		1.960690e-02
4	30	3.412974e-01	0.039049	0.026238	0.033336
True			True		False
1.281096e-02			0.007098		-5.712805e-03
4	40	2.140863e-01	0.021344	0.024339	0.030961
False			True		True
-2.994795e-03			0.006622		9.616717e-03
4	50	1.610155e-01	0.021044	0.018017	0.010608
True			False		False
3.026723e-03			-0.007409		-1.043566e-02
4	60	1.678327e-01	0.010362	0.008650	0.012602
True			True		True
1.712744e-03			0.003952		2.239522e-03
4	70	1.419620e-01	0.009155	0.011251	0.014617
False			True		True
-2.096181e-03			0.003366		5.461920e-03
4	80	1.518022e-01	0.011462	0.010290	0.016220
True			True		True
1.171644e-03			0.005930		4.758071e-03
4	90	1.438736e-01	0.013869	0.009481	0.014771
True			True		True
4.388191e-03			0.005289		9.012930e-04
4	100	1.134666e-01	0.009935	0.007840	0.008615
True			True		False
2.094631e-03			0.000774		-1.320236e-03
4	110	1.040068e-01	0.013745	0.004748	0.008105
True			True		False
8.996873e-03			0.003357		-5.639670e-03
4	120	8.192688e-02	0.008689	0.004786	0.006097
True			True		False
3.903156e-03			0.001311		-2.591889e-03
4	130	9.389321e-02	0.010571	0.005217	0.006681
True			True		False
5.353889e-03			0.001464		-3.889664e-03
4	140	4.544251e-02	0.009376	0.003444	0.005343
True			True		False
5.931820e-03			0.001898		-4.033790e-03
4	150	3.681882e-02	0.006639	0.003390	0.005620
True			True		False
3.249420e-03			0.002230		-1.019445e-03

5	10	7.243675e-01	0.553411	0.864288	0.788894
False			False	True	
-3.108768e-01			-0.075394	2.354824e-01	
5	20	2.998840e-01	0.043677	0.059310	0.018113
False			False	False	
-1.563314e-02			-0.041197	-2.556387e-02	
5	30	2.419078e-01	0.028352	0.042219	0.021004
False			False	False	
-1.386778e-02			-0.021216	-7.347870e-03	
5	40	1.338150e-01	0.021619	0.026061	0.018989
False			False	False	
-4.442077e-03			-0.007072	-2.630388e-03	
5	50	1.288881e-01	0.026912	0.015844	0.019780
True			True	False	
1.106883e-02			0.003936	-7.132641e-03	
5	60	1.178947e-01	0.011379	0.013472	0.006325
False			False	False	
-2.092785e-03			-0.007147	-5.053859e-03	
5	70	1.072224e-01	0.013221	0.014048	0.008327
False			False	False	
-8.271711e-04			-0.005721	-4.893980e-03	
5	80	5.004836e-02	0.009861	0.011314	0.006397
False			False	False	
-1.452737e-03			-0.004916	-3.463546e-03	
5	90	5.434681e-02	0.009221	0.007079	0.007673
True			True	False	
2.142336e-03			0.000594	-1.548340e-03	
5	100	4.447130e-02	0.007371	0.010344	0.007508
False			False	True	
-2.972975e-03			-0.002836	1.372085e-04	
5	110	5.656634e-02	0.009736	0.007947	0.006477
True			False	False	
1.788617e-03			-0.001470	-3.258460e-03	
5	120	5.776641e-02	0.008782	0.008762	0.005527
True			False	False	
1.919127e-05			-0.003235	-3.254086e-03	
5	130	6.123240e-02	0.007603	0.007039	0.005858
True			False	False	
5.635330e-04			-0.001181	-1.744523e-03	
5	140	8.253375e-02	0.009043	0.003007	0.007692
True			True	False	
6.035249e-03			0.004685	-1.350537e-03	
5	150	7.319834e-02	0.006651	0.005362	0.006803
True			True	True	
1.288623e-03			0.001441	1.522875e-04	
6	10	8.051533e-01	0.532609	0.490376	0.711777
True			True	True	
4.223314e-02			0.221400	1.791673e-01	

6	20	3.545966e-01	0.050538	0.037118	0.478545
True			True	True	
1.342052e-02			0.441427	4.280069e-01	
6	30	2.927984e-01	0.026291	0.028245	0.042151
False			True	True	
-1.953512e-03			0.013906	1.585962e-02	
6	40	1.858746e-01	0.027397	0.016981	0.034140
True			True	True	
1.041635e-02			0.017159	6.742292e-03	
6	50	1.557089e-01	0.014702	0.012259	0.011145
True			False	False	
2.443372e-03			-0.001114	-3.557205e-03	
6	60	1.603375e-01	0.013944	0.012933	0.007085
True			False	False	
1.010815e-03			-0.005848	-6.858608e-03	
6	70	1.363430e-01	0.012921	0.010342	0.007837
True			False	False	
2.579545e-03			-0.002504	-5.083713e-03	
6	80	1.267641e-01	0.010229	0.010587	0.007113
False			False	False	
-3.576351e-04			-0.003474	-3.115924e-03	
6	90	1.196828e-01	0.006987	0.009333	0.006033
False			False	False	
-2.346117e-03			-0.003301	-9.544262e-04	
6	100	1.071650e-01	0.005083	0.007224	0.005057
False			False	False	
-2.140498e-03			-0.002167	-2.607357e-05	
6	110	1.016847e-01	0.007881	0.005041	0.003890
True			False	False	
2.840081e-03			-0.001150	-3.990508e-03	
6	120	9.028437e-02	0.007284	0.004994	0.004483
True			False	False	
2.290280e-03			-0.000511	-2.801413e-03	
6	130	8.266053e-02	0.007195	0.005519	0.004332
True			False	False	
1.675489e-03			-0.001187	-2.862908e-03	
6	140	3.063813e-02	0.006095	0.004536	0.004864
True			True	False	
1.559446e-03			0.000328	-1.231038e-03	
6	150	4.558536e-02	0.004990	0.004616	0.005835
True			True	True	
3.737226e-04			0.001219	8.450745e-04	
7	10	7.807441e-01	0.767520	0.589342	0.625018
True			True	False	
1.781783e-01			0.035677	-1.425016e-01	
7	20	3.559138e-01	0.038155	0.025349	0.047584
True			True	True	
1.280688e-02			0.022235	9.428315e-03	

7	30	2.568885e-01	0.021743	0.032888	0.046585
False			True		True
-1.114480e-02			0.013698	2.484245e-02	
7	40	2.091161e-01	0.026051	0.021103	0.022893
True			True		False
4.948661e-03			0.001791	-3.158072e-03	
7	50	1.590896e-01	0.027439	0.012620	0.015474
True			True		False
1.481844e-02			0.002854	-1.196466e-02	
7	60	1.387263e-01	0.016700	0.014726	0.013793
True			False		False
1.974018e-03			-0.000933	-2.906695e-03	
7	70	1.344670e-01	0.012435	0.010034	0.011350
True			True		False
2.401265e-03			0.001316	-1.085725e-03	
7	80	1.423325e-01	0.012618	0.008381	0.007947
True			False		False
4.237577e-03			-0.000434	-4.671462e-03	
7	90	1.470202e-01	0.009633	0.009384	0.006490
True			False		False
2.490183e-04			-0.002893	-3.142408e-03	
7	100	1.408844e-01	0.008594	0.009662	0.005660
False			False		False
-1.067519e-03			-0.004002	-2.934281e-03	
7	110	1.441842e-01	0.009537	0.007950	0.006351
True			False		False
1.587192e-03			-0.001599	-3.185885e-03	
7	120	1.355625e-01	0.006039	0.007019	0.005719
False			False		False
-9.803167e-04			-0.001300	-3.194861e-04	
7	130	1.255816e-01	0.006867	0.009258	0.003869
False			False		False
-2.390962e-03			-0.005389	-2.998501e-03	
7	140	5.543925e-02	0.006443	0.007435	0.004471
False			False		False
-9.919976e-04			-0.002964	-1.971958e-03	
7	150	4.844498e-02	0.010374	0.003598	0.004934
True			True		False
6.775747e-03			0.001336	-5.439901e-03	
8	10	9.859097e-01	0.567022	0.045513	0.829217
True			True		True
5.215087e-01			0.783703	2.621948e-01	
8	20	4.279318e-01	0.036709	0.020227	0.029643
True			True		False
1.648157e-02			0.009416	-7.065396e-03	
8	30	4.162644e-01	0.044326	0.009587	0.035616
True			True		False
3.473949e-02			0.026029	-8.710980e-03	

8	40	2.291338e-01	0.020507	0.014507	0.017176
True			True	False	
6.000007e-03			0.002669	-3.331054e-03	
8	50	2.405370e-01	0.028485	0.016833	0.019620
True			True	False	
1.165235e-02			0.002787	-8.865174e-03	
8	60	2.153741e-01	0.016330	0.008407	0.015071
True			True	False	
7.922674e-03			0.006663	-1.259539e-03	
8	70	1.744268e-01	0.014463	0.004686	0.013021
True			True	False	
9.776839e-03			0.008336	-1.441265e-03	
8	80	1.315993e-01	0.008447	0.006786	0.011346
True			True	True	
1.660904e-03			0.004559	2.898258e-03	
8	90	1.121952e-01	0.009941	0.012190	0.010845
False			False	True	
-2.248923e-03			-0.001345	9.034669e-04	
8	100	1.184332e-01	0.007787	0.010523	0.010537
False			True	True	
-2.736491e-03			0.000014	2.750473e-03	
8	110	8.778156e-02	0.009120	0.009471	0.010347
False			True	True	
-3.513727e-04			0.000876	1.227512e-03	
8	120	9.276900e-02	0.009014	0.006454	0.009604
True			True	True	
2.560378e-03			0.003150	5.896042e-04	
8	130	8.086813e-02	0.007723	0.005254	0.009043
True			True	True	
2.468070e-03			0.003789	1.320844e-03	
8	140	7.862841e-02	0.004405	0.003746	0.008937
True			True	True	
6.585289e-04			0.005190	4.531939e-03	
8	150	5.568683e-02	0.005504	0.003691	0.009321
True			True	True	
1.812853e-03			0.005630	3.817211e-03	
9	10	7.888785e-01	0.105080	0.505788	0.072833
False			False	False	
-4.007081e-01			-0.432955	-3.224698e-02	
9	20	4.292172e-01	0.023762	0.055477	0.041951
False			False	True	
-3.171436e-02			-0.013526	1.818832e-02	
9	30	3.801720e-01	0.017414	0.012722	0.016461
True			True	False	
4.691433e-03			0.003739	-9.525885e-04	
9	40	2.183146e-01	0.011516	0.016549	0.015076
False			False	True	
-5.033904e-03			-0.001473	3.560716e-03	

9	50	1.934871e-01	0.014487	0.010339	0.015362
True			True	True	
4.147399e-03			0.005023	8.755696e-04	
9	60	2.276109e-01	0.011320	0.009170	0.009736
True			True	False	
2.149631e-03			0.000566	-1.584026e-03	
9	70	1.976018e-01	0.012854	0.012625	0.006369
True			False	False	
2.291446e-04			-0.006257	-6.485829e-03	
9	80	1.448053e-01	0.010975	0.012613	0.008418
False			False	False	
-1.637657e-03			-0.004194	-2.556779e-03	
9	90	1.072983e-01	0.011829	0.007304	0.007904
True			True	False	
4.525122e-03			0.000601	-3.924564e-03	
9	100	1.189435e-01	0.004812	0.007164	0.006610
False			False	True	
-2.351518e-03			-0.000554	1.797346e-03	
9	110	9.407970e-02	0.006331	0.006693	0.008247
False			True	True	
-3.624836e-04			0.001554	1.916289e-03	
9	120	7.992638e-02	0.005597	0.005249	0.008171
True			True	True	
3.473691e-04			0.002922	2.574458e-03	
9	130	3.981188e-02	0.006016	0.005147	0.006268
True			True	True	
8.686035e-04			0.001120	2.517525e-04	
9	140	8.879791e-02	0.007369	0.001279	0.004802
True			True	False	
6.090751e-03			0.003523	-2.567344e-03	
9	150	7.493385e-02	0.006733	0.002790	0.004790
True			True	False	
3.943316e-03			0.002000	-1.942952e-03	
10	10	5.242898e-01	0.570494	0.837049	0.787315
False			False	True	
-2.665551e-01			-0.049734	2.168210e-01	
10	20	3.346211e-01	0.034933	0.061001	0.027457
False			False	False	
-2.606866e-02			-0.033544	-7.475466e-03	
10	30	2.926290e-01	0.028060	0.022883	0.015732
True			False	False	
5.177460e-03			-0.007150	-1.232787e-02	
10	40	1.765719e-01	0.011797	0.031690	0.019487
False			False	True	
-1.989240e-02			-0.012203	7.689602e-03	
10	50	1.667657e-01	0.013481	0.017504	0.015874
False			False	True	
-4.023533e-03			-0.001631	2.393031e-03	

10	60	1.417741e-01	0.007517	0.007256	0.011492
True			True	True	
2.601436e-04			0.004236	3.975698e-03	
10	70	1.584835e-01	0.009379	0.006462	0.011551
True			True	True	
2.916490e-03			0.005088	2.171955e-03	
10	80	1.129108e-01	0.006143	0.006642	0.009247
False			True	True	
-4.995253e-04			0.002605	3.104140e-03	
10	90	1.058892e-01	0.008765	0.003979	0.009505
True			True	True	
4.786401e-03			0.005526	7.396549e-04	
10	100	1.177634e-01	0.008755	0.003299	0.009183
True			True	True	
5.455920e-03			0.005884	4.281238e-04	
10	110	1.176572e-01	0.008579	0.003346	0.008680
True			True	True	
5.233349e-03			0.005334	1.003321e-04	
10	120	8.530170e-02	0.009148	0.005362	0.007614
True			True	False	
3.785638e-03			0.002252	-1.533353e-03	
10	130	8.354719e-02	0.007870	0.006559	0.007193
True			True	False	
1.310811e-03			0.000635	-6.761665e-04	
10	140	5.903379e-02	0.007507	0.003812	0.006268
True			True	False	
3.694727e-03			0.002456	-1.238831e-03	
10	150	6.836884e-02	0.005190	0.004493	0.006351
True			True	True	
6.967751e-04			0.001858	1.161052e-03	
11	10	1.757405e+00	0.566914	0.876145	0.534319
False			False	False	
-3.092306e-01			-0.341826	-3.259550e-02	
11	20	2.586236e-01	0.015438	0.663218	0.047584
False			False	True	
-6.477799e-01			-0.615634	3.214589e-02	
11	30	2.074165e-01	0.023006	0.025769	0.041681
False			True	True	
-2.763130e-03			0.015912	1.867490e-02	
11	40	1.438593e-01	0.016499	0.015477	0.019766
True			True	True	
1.021379e-03			0.004289	3.267645e-03	
11	50	9.139687e-02	0.025676	0.008118	0.020250
True			True	False	
1.755766e-02			0.012132	-5.425889e-03	
11	60	9.086268e-02	0.007715	0.012331	0.014718
False			True	True	
-4.616302e-03			0.002387	7.003406e-03	

11	70	9.304319e-02	0.008441	0.010372	0.012301
False			True	True	
-1.930352e-03			0.001929	3.859707e-03	
11	80	1.108135e-01	0.008068	0.015648	0.011284
False			False	True	
-7.579470e-03			-0.004364	3.215964e-03	
11	90	9.944097e-02	0.013536	0.014287	0.009014
False			False	False	
-7.516486e-04			-0.005273	-4.521787e-03	
11	100	8.605010e-02	0.014110	0.010747	0.009975
True			False	False	
3.362629e-03			-0.000772	-4.135078e-03	
11	110	6.795818e-02	0.012016	0.009286	0.008471
True			False	False	
2.730125e-03			-0.000815	-3.544842e-03	
11	120	7.791186e-02	0.009660	0.005567	0.009197
True			True	False	
4.092982e-03			0.003630	-4.627147e-04	
11	130	6.991798e-02	0.007813	0.008906	0.009296
False			True	True	
-1.093127e-03			0.000391	1.483702e-03	
11	140	1.040905e-01	0.006731	0.005965	0.008063
True			True	True	
7.660331e-04			0.002098	1.332263e-03	
11	150	9.213711e-02	0.005688	0.005574	0.007223
True			True	True	
1.146501e-04			0.001650	1.534860e-03	
12	10	6.479964e-01	0.106396	0.100799	0.769966
True			True	True	
5.597053e-03			0.669167	6.635702e-01	
12	20	5.172537e-01	0.049112	0.071101	0.657469
False			True	True	
-2.198926e-02			0.586368	6.083575e-01	
12	30	2.335863e-01	0.051389	0.040062	0.017613
True			False	False	
1.132732e-02			-0.022449	-3.377602e-02	
12	40	9.699654e-02	0.025119	0.029366	0.013010
False			False	False	
-4.247730e-03			-0.016357	-1.210902e-02	
12	50	1.327177e-01	0.011724	0.025049	0.016932
False			False	True	
-1.332514e-02			-0.008117	5.208449e-03	
12	60	7.216192e-02	0.017984	0.013614	0.012157
True			False	False	
4.369540e-03			-0.001458	-5.827173e-03	
12	70	6.922369e-02	0.013184	0.011204	0.013835
True			True	True	
1.980352e-03			0.002631	6.507611e-04	

12	80	9.632592e-02	0.009196	0.008637	0.010108
True			True	True	
5.585281e-04			0.001471	9.122049e-04	
12	90	8.393149e-02	0.010876	0.008609	0.010597
True			True	False	
2.266222e-03			0.001988	-2.782799e-04	
12	100	1.120108e-01	0.010134	0.009411	0.009577
True			True	False	
7.230531e-04			0.000167	-5.564254e-04	
12	110	1.081272e-01	0.010204	0.010425	0.008258
False			False	False	
-2.208548e-04			-0.002167	-1.945834e-03	
12	120	8.757364e-02	0.008525	0.006832	0.006517
True			False	False	
1.693578e-03			-0.000315	-2.008391e-03	
12	130	8.809724e-02	0.005320	0.004294	0.006943
True			True	True	
1.026747e-03			0.002649	1.622269e-03	
12	140	4.058261e-02	0.004550	0.004311	0.006404
True			True	True	
2.392080e-04			0.002093	1.853545e-03	
12	150	4.747523e-02	0.002594	0.006952	0.006372
False			False	True	
-4.357852e-03			-0.000580	3.777835e-03	
13	10	7.724040e-01	0.750616	0.663653	0.626775
True			False	False	
8.696327e-02			-0.036878	-1.238416e-01	
13	20	2.497808e-01	0.055430	0.030910	0.108454
True			True	True	
2.452036e-02			0.077544	5.302393e-02	
13	30	1.904051e-01	0.019232	0.055431	0.017701
False			False	False	
-3.619900e-02			-0.037730	-1.531016e-03	
13	40	2.070238e-01	0.029197	0.031606	0.014773
False			False	False	
-2.408257e-03			-0.016833	-1.442463e-02	
13	50	1.836491e-01	0.026168	0.019997	0.016868
True			False	False	
6.170940e-03			-0.003129	-9.299981e-03	
13	60	1.394802e-01	0.013114	0.011110	0.014055
True			True	True	
2.004504e-03			0.002946	9.413843e-04	
13	70	1.401842e-01	0.011607	0.003055	0.011989
True			True	True	
8.552989e-03			0.008934	3.811837e-04	
13	80	1.637255e-01	0.010292	0.007134	0.008787
True			True	False	
3.158682e-03			0.001654	-1.505168e-03	

13	90	1.417704e-01	0.009588	0.005734	0.006940
True			True	False	
3.853563e-03			0.001206	-2.647714e-03	
13	100	1.190306e-01	0.007421	0.005052	0.004939
True			False	False	
2.369314e-03			-0.000112	-2.481741e-03	
13	110	8.651719e-02	0.005729	0.007559	0.004180
False			False	False	
-1.829481e-03			-0.003378	-1.548973e-03	
13	120	8.157123e-02	0.005448	0.006361	0.006094
False			False	True	
-9.130338e-04			-0.000267	6.459543e-04	
13	130	8.933229e-02	0.004646	0.005348	0.006651
False			True	True	
-7.014010e-04			0.001303	2.004366e-03	
13	140	6.372151e-02	0.004612	0.004727	0.006128
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-1.152061e-04			0.001401	1.516125e-03	
13	150	5.302015e-02	0.005340	0.005030	0.005377
True			True	True	
3.096841e-04			0.000347	3.737794e-05	
14	10	4.614227e-01	0.807686	0.089633	0.468141
True			True	False	
7.180534e-01			0.378509	-3.395448e-01	
14	20	2.595480e-01	0.038155	0.035344	0.052260
True			True	True	
2.811679e-03			0.016917	1.410487e-02	
14	30	1.990943e-01	0.041218	0.029042	0.032937
True			True	False	
1.217604e-02			0.003895	-8.280797e-03	
14	40	2.357505e-01	0.022532	0.010739	0.021198
True			True	False	
1.179279e-02			0.010459	-1.333580e-03	
14	50	1.964386e-01	0.008062	0.009578	0.016641
False			True	True	
-1.515633e-03			0.007063	8.578490e-03	
14	60	1.552161e-01	0.017576	0.017240	0.011790
True			False	False	
3.361063e-04			-0.005449	-5.785593e-03	
14	70	1.703201e-01	0.017467	0.016160	0.009316
True			False	False	
1.307320e-03			-0.006845	-8.151938e-03	
14	80	1.686456e-01	0.014501	0.006039	0.009291
True			True	False	
8.462478e-03			0.003252	-5.210494e-03	
14	90	1.727219e-01	0.009511	0.006984	0.009391
True			True	False	
2.526602e-03			0.002406	-1.202222e-04	

14	100	1.365449e-01	0.006114	0.009094	0.009266
False			True		True
-2.980005e-03			0.000172	3.151875e-03	
14	110	1.336880e-01	0.008163	0.005288	0.010237
True			True		True
2.875417e-03			0.004950	2.074184e-03	
14	120	1.223905e-01	0.006736	0.007007	0.009991
False			True		True
-2.712172e-04			0.002984	3.255179e-03	
14	130	1.120375e-01	0.005817	0.004491	0.009831
True			True		True
1.326668e-03			0.005340	4.013360e-03	
14	140	6.921406e-02	0.008492	0.004574	0.009059
True			True		True
3.918314e-03			0.004485	5.670111e-04	
14	150	6.023914e-02	0.004416	0.004274	0.007817
True			True		True
1.423725e-04			0.003543	3.400491e-03	
15	10	8.003523e-01	0.828676	0.826534	0.766588
True			False		False
2.142359e-03			-0.059945	-6.208774e-02	
15	20	3.408729e-01	0.082535	0.080091	0.041288
True			False		False
2.443954e-03			-0.038803	-4.124652e-02	
15	30	2.338454e-01	0.026276	0.033338	0.034056
False			True		True
-7.061970e-03			0.000718	7.780300e-03	
15	40	1.585964e-01	0.038455	0.025877	0.023975
True			False		False
1.257881e-02			-0.001902	-1.448051e-02	
15	50	1.540821e-01	0.019971	0.019510	0.020188
True			True		True
4.614561e-04			0.000678	2.165262e-04	
15	60	1.362510e-01	0.015330	0.018493	0.015492
False			False		True
-3.162878e-03			-0.003001	1.617041e-04	
15	70	1.290598e-01	0.018932	0.016007	0.014600
True			False		False
2.924867e-03			-0.001406	-4.331267e-03	
15	80	1.115124e-01	0.013448	0.005228	0.009128
True			True		False
8.219653e-03			0.003900	-4.319990e-03	
15	90	9.196018e-02	0.010123	0.014259	0.008454
False			False		False
-4.135945e-03			-0.005804	-1.668331e-03	
15	100	1.201370e-01	0.011598	0.007269	0.007988
True			True		False
4.329695e-03			0.000720	-3.609984e-03	

15	110	1.077214e-01	0.004325	0.005935	0.008198
False			True	True	
-1.609917e-03			0.002263	3.872450e-03	
15	120	9.215947e-02	0.005700	0.007787	0.006339
False			False	True	
-2.086742e-03			-0.001448	6.389724e-04	
15	130	9.833264e-02	0.001712	0.004743	0.004683
False			False	True	
-3.030740e-03			-0.000060	2.970366e-03	
15	140	6.615501e-02	0.003126	0.004574	0.005286
False			True	True	
-1.448444e-03			0.000712	2.160387e-03	
15	150	6.178479e-02	0.006409	0.005660	0.005036
True			False	False	
7.487917e-04			-0.000625	-1.373586e-03	
16	10	6.916960e-01	0.708113	0.769506	0.101125
False			False	False	
-6.139311e-02			-0.668382	-6.069886e-01	
16	20	2.982446e-01	0.066978	0.078121	0.059486
False			False	False	
-1.114227e-02			-0.018635	-7.492800e-03	
16	30	3.614489e-01	0.035365	0.036406	0.026039
False			False	False	
-1.040968e-03			-0.010367	-9.326198e-03	
16	40	1.941402e-01	0.024700	0.014874	0.027029
True			True	True	
9.826416e-03			0.012156	2.329308e-03	
16	50	1.581808e-01	0.009010	0.011533	0.025698
False			True	True	
-2.522164e-03			0.014166	1.668799e-02	
16	60	1.357986e-01	0.013788	0.015614	0.013184
False			False	False	
-1.825970e-03			-0.002430	-6.036902e-04	
16	70	1.349774e-01	0.008284	0.008486	0.011905
False			True	True	
-2.023108e-04			0.003419	3.621131e-03	
16	80	1.245433e-01	0.006985	0.007029	0.013492
False			True	True	
-4.403280e-05			0.006463	6.507249e-03	
16	90	1.324695e-01	0.003363	0.004874	0.010671
False			True	True	
-1.511736e-03			0.005797	7.308622e-03	
16	100	1.020188e-01	0.004618	0.005369	0.009504
False			True	True	
-7.512144e-04			0.004134	4.885667e-03	
16	110	9.668226e-02	0.005546	0.006501	0.008317
False			True	True	
-9.548435e-04			0.001816	2.771138e-03	

16	120	7.852144e-02	0.004049	0.002159	0.009741
True			True	True	
1.890562e-03			0.007582	5.691207e-03	
16	130	8.666225e-02	0.005140	0.003080	0.008454
True			True	True	
2.060545e-03			0.005374	3.313857e-03	
16	140	5.264380e-02	0.005173	0.005243	0.008406
False			True	True	
-6.942817e-05			0.003163	3.232398e-03	
16	150	4.434293e-02	0.004883	0.003400	0.008098
True			True	True	
1.482562e-03			0.004698	3.215037e-03	
17	10	4.920912e-01	0.821776	0.836820	0.734358
False			False	False	
-1.504422e-02			-0.102462	-8.741789e-02	
17	20	5.620023e-01	0.034376	0.342303	0.025551
False			False	False	
-3.079274e-01			-0.316752	-8.824513e-03	
17	30	3.448604e-01	0.021947	0.042456	0.028163
False			False	True	
-2.050889e-02			-0.014293	6.216268e-03	
17	40	2.870421e-01	0.021015	0.014500	0.034485
True			True	True	
6.514759e-03			0.019984	1.346957e-02	
17	50	1.603822e-01	0.019180	0.020975	0.026473
False			True	True	
-1.794894e-03			0.005498	7.293297e-03	
17	60	1.603687e-01	0.015349	0.011214	0.014605
True			True	False	
4.135126e-03			0.003391	-7.438647e-04	
17	70	1.572397e-01	0.012761	0.014676	0.007018
False			False	False	
-1.915066e-03			-0.007658	-5.742770e-03	
17	80	1.465738e-01	0.012943	0.010451	0.005211
True			False	False	
2.492448e-03			-0.005240	-7.732503e-03	
17	90	1.625678e-01	0.006854	0.009011	0.006881
False			False	True	
-2.157380e-03			-0.002131	2.658729e-05	
17	100	1.438818e-01	0.007601	0.014105	0.006957
False			False	False	
-6.503610e-03			-0.007148	-6.439197e-04	
17	110	1.237758e-01	0.007897	0.008592	0.006096
False			False	False	
-6.945209e-04			-0.002496	-1.801095e-03	
17	120	9.972300e-02	0.007166	0.006185	0.005013
True			False	False	
9.812949e-04			-0.001172	-2.153398e-03	

17	130	8.153701e-02	0.008313	0.004604	0.004694
True			True	False	
3.708888e-03			0.000090	-3.619067e-03	
17	140	8.062664e-02	0.008466	0.003772	0.003222
True			False	False	
4.694275e-03			-0.000550	-5.244407e-03	
17	150	6.973752e-02	0.007762	0.007485	0.002861
True			False	False	
2.767490e-04			-0.004624	-4.900989e-03	
18	10	1.741993e+00	0.117030	0.110083	0.061504
True			False	False	
6.946769e-03			-0.048579	-5.552561e-02	
18	20	4.547182e-01	0.057234	0.026298	0.112768
True			True	True	
3.093562e-02			0.086470	5.553442e-02	
18	30	2.620980e-01	0.020748	0.018728	0.039156
True			True	True	
2.020609e-03			0.020428	1.840785e-02	
18	40	2.266596e-01	0.015437	0.019944	0.016549
False			False	True	
-4.506305e-03			-0.003395	1.111278e-03	
18	50	2.319452e-01	0.014270	0.015639	0.016537
False			True	True	
-1.368873e-03			0.000898	2.267141e-03	
18	60	1.926882e-01	0.019534	0.015871	0.008507
True			False	False	
3.663687e-03			-0.007363	-1.102681e-02	
18	70	1.401084e-01	0.012605	0.006161	0.010186
True			True	False	
6.444471e-03			0.004025	-2.419308e-03	
18	80	1.189565e-01	0.011037	0.005273	0.012476
True			True	True	
5.763386e-03			0.007202	1.438988e-03	
18	90	1.010084e-01	0.009270	0.005451	0.012768
True			True	True	
3.819351e-03			0.007318	3.498323e-03	
18	100	7.820294e-02	0.006958	0.009398	0.011851
False			True	True	
-2.440629e-03			0.002453	4.893766e-03	
18	110	7.550906e-02	0.006381	0.007020	0.010718
False			True	True	
-6.386442e-04			0.003698	4.336893e-03	
18	120	8.029640e-02	0.005916	0.006357	0.008257
False			True	True	
-4.413486e-04			0.001900	2.340982e-03	
18	130	6.576266e-02	0.005161	0.008654	0.007691
False			False	True	
-3.492840e-03			-0.000962	2.530544e-03	

18	140	6.487880e-02	0.005102	0.008777	0.006595
False			False	True	
-3.674912e-03			-0.002181	1.493574e-03	
18	150	4.779077e-02	0.005638	0.005906	0.006223
False			True	True	
-2.682689e-04			0.000317	5.851830e-04	
19	10	1.388237e+00	0.713776	0.110217	0.771159
True			True	True	
6.035583e-01			0.660941	5.738303e-02	
19	20	5.243061e-01	0.081246	0.066651	0.037474
True			False	False	
1.459485e-02			-0.029177	-4.377192e-02	
19	30	5.611621e-01	0.051974	0.066651	0.022762
False			False	False	
-1.467717e-02			-0.043889	-2.921143e-02	
19	40	2.188045e-01	0.018026	0.074135	0.014538
False			False	False	
-5.610842e-02			-0.059597	-3.488484e-03	
19	50	2.043732e-01	0.020087	0.026892	0.007229
False			False	False	
-6.804306e-03			-0.019662	-1.285815e-02	
19	60	1.885879e-01	0.019209	0.015487	0.015910
True			True	False	
3.722419e-03			0.000423	-3.299805e-03	
19	70	1.740339e-01	0.010883	0.011770	0.015878
False			True	True	
-8.865381e-04			0.004108	4.994183e-03	
19	80	1.705181e-01	0.009030	0.010269	0.015274
False			True	True	
-1.239405e-03			0.005004	6.243766e-03	
19	90	1.641246e-01	0.008410	0.007863	0.016093
True			True	True	
5.467172e-04			0.008230	7.682862e-03	
19	100	1.342278e-01	0.008583	0.010225	0.012340
False			True	True	
-1.642044e-03			0.002115	3.756936e-03	
19	110	1.316581e-01	0.008435	0.006759	0.011952
True			True	True	
1.675626e-03			0.005193	3.517466e-03	
19	120	1.212242e-01	0.004998	0.008516	0.010528
False			True	True	
-3.517822e-03			0.002012	5.529759e-03	
19	130	1.095304e-01	0.005995	0.008447	0.011179
False			True	True	
-2.452215e-03			0.002732	5.184039e-03	
19	140	8.293665e-02	0.006407	0.006307	0.011566
True			True	True	
9.938941e-05			0.005258	5.158668e-03	

19	150	7.564134e-02	0.007862	0.005515	0.011186
True			True	True	
2.347090e-03			0.005671	3.324191e-03	
20	10	1.184615e+00	0.706111	0.822277	0.110797
False			False	False	
-1.161662e-01			-0.711480	-5.953141e-01	
20	20	7.884979e-01	0.031751	0.058198	0.045030
False			False	True	
-2.644729e-02			-0.013168	1.327892e-02	
20	30	4.133436e-01	0.023323	0.055431	0.027437
False			False	True	
-3.210788e-02			-0.027993	4.114665e-03	
20	40	2.303626e-01	0.037307	0.018624	0.021139
True			True	False	
1.868348e-02			0.002515	-1.616817e-02	
20	50	2.292581e-01	0.005593	0.015861	0.016979
False			True	True	
-1.026816e-02			0.001118	1.138635e-02	
20	60	2.033697e-01	0.012003	0.014669	0.016900
False			True	True	
-2.665867e-03			0.002231	4.896848e-03	
20	70	1.809489e-01	0.010922	0.008878	0.013439
True			True	True	
2.043978e-03			0.004561	2.517026e-03	
20	80	1.422910e-01	0.013723	0.007901	0.010110
True			True	False	
5.822080e-03			0.002208	-3.613785e-03	
20	90	1.415577e-01	0.011591	0.013286	0.011626
False			False	True	
-1.695225e-03			-0.001661	3.451946e-05	
20	100	1.064902e-01	0.008967	0.012725	0.010588
False			False	True	
-3.758266e-03			-0.002137	1.620858e-03	
20	110	8.996087e-02	0.010747	0.010592	0.007947
True			False	False	
1.548388e-04			-0.002645	-2.800015e-03	
20	120	8.371130e-02	0.008409	0.003584	0.006391
True			True	False	
4.825788e-03			0.002807	-2.018315e-03	
20	130	7.054136e-02	0.008076	0.004930	0.006505
True			True	False	
3.145824e-03			0.001575	-1.570664e-03	
20	140	8.729850e-02	0.008233	0.005200	0.005630
True			True	False	
3.032680e-03			0.000430	-2.602305e-03	
20	150	5.148519e-02	0.006742	0.002420	0.003814
True			True	False	
4.322034e-03			0.001394	-2.927970e-03	

21	10	4.699133e-01	0.101742	0.101742	0.814813
True			True	True	
2.512981e-08			0.713071	7.130711e-01	
21	20	3.328006e-01	0.037872	0.038587	0.701133
False			True	True	
-7.158345e-04			0.662545	6.632612e-01	
21	30	2.768146e-01	0.040120	0.025779	0.030061
True			True	False	
1.434109e-02			0.004282	-1.005901e-02	
21	40	1.858652e-01	0.031633	0.029433	0.026588
True			False	False	
2.200385e-03			-0.002844	-5.044793e-03	
21	50	2.120410e-01	0.020960	0.017447	0.017478
True			True	False	
3.512272e-03			0.000031	-3.481219e-03	
21	60	1.486163e-01	0.013739	0.017058	0.014068
False			False	True	
-3.318837e-03			-0.002989	3.294323e-04	
21	70	1.522865e-01	0.014613	0.016027	0.014530
False			False	False	
-1.413884e-03			-0.001497	-8.307185e-05	
21	80	1.514686e-01	0.013841	0.012898	0.010624
True			False	False	
9.434291e-04			-0.002273	-3.216927e-03	
21	90	1.627105e-01	0.012501	0.012167	0.011947
True			False	False	
3.338250e-04			-0.000220	-5.543186e-04	
21	100	1.396976e-01	0.011574	0.010054	0.011253
True			True	False	
1.519755e-03			0.001199	-3.210717e-04	
21	110	1.434927e-01	0.011324	0.009948	0.010139
True			True	False	
1.375734e-03			0.000191	-1.184380e-03	
21	120	1.337360e-01	0.010386	0.005470	0.009141
True			True	False	
4.915399e-03			0.003671	-1.244763e-03	
21	130	1.108440e-01	0.008913	0.005450	0.007060
True			True	False	
3.462534e-03			0.001610	-1.852974e-03	
21	140	9.022725e-02	0.009400	0.004879	0.006495
True			True	False	
4.520963e-03			0.001615	-2.905501e-03	
21	150	8.858510e-02	0.009888	0.004037	0.006144
True			True	False	
5.851245e-03			0.002108	-3.743662e-03	
22	10	4.165461e-01	0.695586	0.853727	0.760948
False			False	True	
-1.581404e-01			-0.092778	6.536216e-02	

22	20	4.758001e-01	0.084036	0.560481	0.062442
False			False		False
-4.764451e-01			-0.498039	-2.159409e-02	
22	30	3.452236e-01	0.048177	0.026902	0.036829
True			True		False
2.127571e-02			0.009928	-1.134802e-02	
22	40	1.726543e-01	0.019927	0.024014	0.036006
False			True		True
-4.086976e-03			0.011992	1.607888e-02	
22	50	1.513391e-01	0.017109	0.011897	0.036131
True			True		True
5.211089e-03			0.024233	1.902227e-02	
22	60	1.202005e-01	0.013009	0.010570	0.008426
True			False		False
2.439167e-03			-0.002143	-4.582478e-03	
22	70	9.534384e-02	0.013932	0.017111	0.009149
False			False		False
-3.179516e-03			-0.007962	-4.782845e-03	
22	80	1.238000e-01	0.013083	0.011146	0.010162
True			False		False
1.937284e-03			-0.000984	-2.921361e-03	
22	90	8.514021e-02	0.008781	0.010174	0.009406
False			False		True
-1.393799e-03			-0.000768	6.253308e-04	
22	100	9.815954e-02	0.010020	0.010346	0.007850
False			False		False
-3.259912e-04			-0.002497	-2.170612e-03	
22	110	7.096244e-02	0.006692	0.006839	0.008690
False			True		True
-1.472037e-04			0.001851	1.998032e-03	
22	120	6.967552e-02	0.006330	0.006799	0.009030
False			True		True
-4.690446e-04			0.002231	2.700112e-03	
22	130	7.782167e-02	0.004754	0.008016	0.008699
False			True		True
-3.261967e-03			0.000684	3.945751e-03	
22	140	5.495207e-02	0.006040	0.004508	0.008154
True			True		True
1.531722e-03			0.003646	2.114705e-03	
22	150	5.382414e-02	0.007825	0.004224	0.007598
True			True		False
3.600770e-03			0.003373	-2.273873e-04	
23	10	1.764531e+00	0.089277	0.836810	0.697600
False			False		True
-7.475329e-01			-0.139209	6.083237e-01	
23	20	2.672352e-01	0.066651	0.076769	0.032439
False			False		False
-1.011791e-02			-0.044330	-3.421201e-02	

23	30	1.710803e-01	0.016860	0.039675	0.043903
False			True	True	
-2.281544e-02			0.004228	2.704346e-02	
23	40	1.414656e-01	0.021225	0.019783	0.030899
True			True	True	
1.442610e-03			0.011117	9.674266e-03	
23	50	1.015833e-01	0.027896	0.033937	0.019765
False			False	False	
-6.041942e-03			-0.014172	-8.130197e-03	
23	60	9.580892e-02	0.011363	0.017765	0.015021
False			False	True	
-6.402326e-03			-0.002743	3.658918e-03	
23	70	9.255755e-02	0.010786	0.017199	0.007665
False			False	False	
-6.412861e-03			-0.009535	-3.121685e-03	
23	80	9.113704e-02	0.016672	0.007964	0.008374
True			True	False	
8.707406e-03			0.000410	-8.297405e-03	
23	90	9.615446e-02	0.005938	0.012869	0.009122
False			False	True	
-6.931044e-03			-0.003747	3.184345e-03	
23	100	6.994440e-02	0.005434	0.006766	0.010463
False			True	True	
-1.332171e-03			0.003697	5.028976e-03	
23	110	7.347990e-02	0.007716	0.005587	0.010164
True			True	True	
2.128593e-03			0.004577	2.448007e-03	
23	120	7.075201e-02	0.006634	0.008876	0.009848
False			True	True	
-2.242555e-03			0.000972	3.214192e-03	
23	130	6.245304e-02	0.007115	0.007496	0.009209
False			True	True	
-3.806135e-04			0.001714	2.094189e-03	
23	140	6.865662e-02	0.007223	0.009303	0.009230
False			False	True	
-2.080615e-03			-0.000074	2.006937e-03	
23	150	7.025316e-02	0.008472	0.006577	0.007354
True			True	False	
1.894973e-03			0.000777	-1.118347e-03	
24	10	7.692112e-01	0.475548	0.475548	0.104567
True			False	False	
3.885781e-16			-0.370981	-3.709807e-01	
24	20	2.266735e-01	0.090561	0.098185	0.040811
False			False	False	
-7.623697e-03			-0.057374	-4.975015e-02	
24	30	3.211148e-01	0.021539	0.025644	0.023277
False			False	True	
-4.105002e-03			-0.002368	1.737351e-03	

24	40	2.304143e-01	0.018525	0.021584	0.014804
False			False		False
-3.059327e-03			-0.006780	-3.720527e-03	
24	50	1.698158e-01	0.016519	0.014267	0.013278
True			False		False
2.251671e-03			-0.000989	-3.241074e-03	
24	60	1.545873e-01	0.017748	0.010544	0.007082
True			False		False
7.203592e-03			-0.003462	-1.066575e-02	
24	70	1.383625e-01	0.011897	0.011115	0.007549
True			False		False
7.817179e-04			-0.003566	-4.347859e-03	
24	80	1.219656e-01	0.012580	0.009979	0.005962
True			False		False
2.600459e-03			-0.004017	-6.617617e-03	
24	90	1.236714e-01	0.010087	0.008342	0.006225
True			False		False
1.745289e-03			-0.002117	-3.862325e-03	
24	100	8.233800e-02	0.008809	0.006055	0.005742
True			False		False
2.754106e-03			-0.000313	-3.067321e-03	
24	110	8.340907e-02	0.008616	0.003965	0.005180
True			True		False
4.650853e-03			0.001215	-3.435495e-03	
24	120	6.271898e-02	0.007551	0.006407	0.005610
True			False		False
1.143651e-03			-0.000797	-1.940822e-03	
24	130	5.127760e-02	0.006873	0.007810	0.004685
False			False		False
-9.366597e-04			-0.003125	-2.188638e-03	
24	140	6.940784e-02	0.005504	0.006438	0.004640
False			False		False
-9.335591e-04			-0.001798	-8.647670e-04	
24	150	6.137964e-02	0.004311	0.004641	0.004222
False			False		False
-3.295078e-04			-0.000419	-8.926878e-05	
25	10	1.184705e+00	0.582005	0.851474	0.597040
False			False		True
-2.694694e-01			-0.254434	1.503491e-02	
25	20	5.990716e-01	0.048809	0.585838	0.024965
False			False		False
-5.370290e-01			-0.560873	-2.384365e-02	
25	30	3.882558e-01	0.049432	0.213510	0.020499
False			False		False
-1.640776e-01			-0.193011	-2.893329e-02	
25	40	1.436393e-01	0.022502	0.026854	0.013643
False			False		False
-4.352002e-03			-0.013211	-8.858603e-03	

25	50	9.753036e-02	0.021853	0.022231	0.014206
False			False		False
-3.782325e-04			-0.008026	-7.647357e-03	
25	60	6.010456e-02	0.015768	0.021664	0.007340
False			False		False
-5.896226e-03			-0.014324	-8.428123e-03	
25	70	8.422200e-02	0.011108	0.010412	0.006550
True			False		False
6.953971e-04			-0.003862	-4.557733e-03	
25	80	1.075878e-01	0.007938	0.015203	0.007579
False			False		False
-7.264945e-03			-0.007624	-3.594272e-04	
25	90	1.037997e-01	0.010715	0.004048	0.008022
True			True		False
6.667472e-03			0.003974	-2.693112e-03	
25	100	9.830672e-02	0.010783	0.005057	0.006954
True			True		False
5.726132e-03			0.001898	-3.828317e-03	
25	110	9.676389e-02	0.010956	0.004191	0.006041
True			True		False
6.764063e-03			0.001850	-4.914365e-03	
25	120	8.535300e-02	0.008940	0.003636	0.006858
True			True		False
5.303987e-03			0.003222	-2.081708e-03	
25	130	6.337590e-02	0.006594	0.004706	0.002772
True			False		False
1.888135e-03			-0.001934	-3.821777e-03	
25	140	9.890170e-02	0.007219	0.003678	0.002429
True			False		False
3.541487e-03			-0.001249	-4.790585e-03	
25	150	8.291859e-02	0.005422	0.003916	0.003477
True			False		False
1.505662e-03			-0.000438	-1.944086e-03	
26	10	5.803946e-01	0.780008	0.796971	0.791678
False			False		True
-1.696236e-02			-0.005292	1.167010e-02	
26	20	3.540707e-01	0.707825	0.041099	0.708770
True			True		True
6.667264e-01			0.667671	9.446948e-04	
26	30	2.574641e-01	0.055431	0.049052	0.026198
True			False		False
6.378458e-03			-0.022854	-2.923236e-02	
26	40	2.754633e-01	0.010617	0.019899	0.023247
False			True		True
-9.282868e-03			0.003347	1.263034e-02	
26	50	2.533356e-01	0.009226	0.015643	0.023802
False			True		True
-6.417562e-03			0.008159	1.457648e-02	

26	60	2.481244e-01	0.020229	0.017482	0.009554
True			False		False
2.746996e-03			-0.007928	-1.067513e-02	
26	70	1.841623e-01	0.019661	0.018907	0.008522
True			False		False
7.547185e-04			-0.010385	-1.113981e-02	
26	80	1.587553e-01	0.015869	0.015527	0.008655
True			False		False
3.425798e-04			-0.006872	-7.214256e-03	
26	90	1.425341e-01	0.011366	0.007747	0.008138
True			True		False
3.618374e-03			0.000391	-3.227435e-03	
26	100	1.191871e-01	0.009020	0.006597	0.007148
True			True		False
2.423466e-03			0.000551	-1.872411e-03	
26	110	1.158721e-01	0.008542	0.005934	0.006255
True			True		False
2.607314e-03			0.000320	-2.287012e-03	
26	120	9.922687e-02	0.008020	0.007291	0.005239
True			False		False
7.292052e-04			-0.002051	-2.780519e-03	
26	130	9.170856e-02	0.008382	0.004153	0.005968
True			True		False
4.228577e-03			0.001815	-2.413937e-03	
26	140	6.412208e-02	0.007180	0.004180	0.005401
True			True		False
2.999888e-03			0.001221	-1.778844e-03	
26	150	5.026679e-02	0.004041	0.005496	0.005301
False			False		True
-1.454883e-03			-0.000195	1.260046e-03	
27	10	5.917277e-01	0.704101	0.438233	0.047532
True			False		False
2.658682e-01			-0.390701	-6.565691e-01	
27	20	2.814359e-01	0.035727	0.025447	0.038604
True			True		True
1.028069e-02			0.013158	2.877015e-03	
27	30	2.329164e-01	0.032232	0.035078	0.023155
False			False		False
-2.846682e-03			-0.011923	-9.076500e-03	
27	40	2.266073e-01	0.012627	0.011306	0.017173
True			True		True
1.321138e-03			0.005867	4.545828e-03	
27	50	2.297928e-01	0.017994	0.022117	0.012425
False			False		False
-4.123401e-03			-0.009692	-5.568311e-03	
27	60	2.052578e-01	0.010690	0.011868	0.012282
False			True		True
-1.178523e-03			0.000414	1.592467e-03	

27	70	1.870604e-01	0.004965	0.006445	0.011875
False			True		True
-1.479872e-03			0.005430	6.909769e-03	
27	80	1.641815e-01	0.006339	0.009975	0.010705
False			True		True
-3.635918e-03			0.000730	4.365797e-03	
27	90	1.284609e-01	0.010139	0.011029	0.009976
False			False		False
-8.899222e-04			-0.001053	-1.631222e-04	
27	100	1.002645e-01	0.009560	0.009522	0.010209
True			True		True
3.857447e-05			0.000687	6.483830e-04	
27	110	9.669936e-02	0.008601	0.007307	0.009868
True			True		True
1.294306e-03			0.002562	1.267267e-03	
27	120	5.858563e-02	0.007834	0.005270	0.009150
True			True		True
2.564314e-03			0.003880	1.315809e-03	
27	130	6.474475e-02	0.004052	0.003096	0.008481
True			True		True
9.557148e-04			0.005385	4.429219e-03	
27	140	7.444301e-02	0.003778	0.005994	0.008597
False			True		True
-2.216008e-03			0.002603	4.819280e-03	
27	150	5.123811e-02	0.005116	0.003085	0.007612
True			True		True
2.030418e-03			0.004526	2.495631e-03	
28	10	6.657360e-01	0.559911	0.789827	0.477103
False			False		False
-2.299163e-01			-0.312725	-8.280844e-02	
28	20	3.830594e-01	0.031436	0.651373	0.045347
False			False		True
-6.199372e-01			-0.606026	1.391067e-02	
28	30	3.364329e-01	0.050534	0.021224	0.023807
True			True		False
2.931011e-02			0.002582	-2.672771e-02	
28	40	2.137257e-01	0.048149	0.651373	0.021534
False			False		False
-6.032244e-01			-0.629839	-2.661477e-02	
28	50	2.359279e-01	0.010736	0.015475	0.018564
False			True		True
-4.739492e-03			0.003089	7.828076e-03	
28	60	1.603260e-01	0.014713	0.011188	0.015169
True			True		True
3.525104e-03			0.003981	4.559808e-04	
28	70	1.344009e-01	0.011572	0.005044	0.015995
True			True		True
6.527806e-03			0.010951	4.423651e-03	

28	80	9.113212e-02	0.010927	0.007698	0.010835
True			True	False	
3.228880e-03			0.003137	-9.200605e-05	
28	90	7.479165e-02	0.009913	0.006314	0.009716
True			True	False	
3.598967e-03			0.003402	-1.965189e-04	
28	100	9.574054e-02	0.006890	0.004953	0.011205
True			True	True	
1.937591e-03			0.006252	4.314609e-03	
28	110	8.719060e-02	0.005622	0.004432	0.009504
True			True	True	
1.190388e-03			0.005073	3.882196e-03	
28	120	6.258515e-02	0.006828	0.005952	0.011192
True			True	True	
8.758900e-04			0.005240	4.364311e-03	
28	130	6.540637e-02	0.006949	0.006443	0.010917
True			True	True	
5.068820e-04			0.004475	3.967812e-03	
28	140	3.836788e-02	0.007050	0.002968	0.009798
True			True	True	
4.081112e-03			0.006830	2.748702e-03	
28	150	3.602914e-02	0.005257	0.004201	0.009338
True			True	True	
1.055841e-03			0.005137	4.080836e-03	
29	10	5.803944e-01	0.775667	0.891397	0.490056
False			False	False	
-1.157295e-01			-0.401341	-2.856116e-01	
29	20	3.994619e-01	0.046313	0.711368	0.043622
False			False	False	
-6.650552e-01			-0.667746	-2.690524e-03	
29	30	2.296086e-01	0.048661	0.047305	0.042001
True			False	False	
1.355126e-03			-0.005305	-6.659950e-03	
29	40	2.326958e-01	0.015376	0.041054	0.044767
False			True	True	
-2.567739e-02			0.003714	2.939098e-02	
29	50	2.391267e-01	0.006969	0.017098	0.034400
False			True	True	
-1.012955e-02			0.017301	2.743056e-02	
29	60	1.622027e-01	0.012392	0.015209	0.013516
False			False	True	
-2.816887e-03			-0.001694	1.123335e-03	
29	70	1.244490e-01	0.011135	0.008541	0.013499
True			True	True	
2.594111e-03			0.004958	2.364002e-03	
29	80	8.681017e-02	0.004763	0.009681	0.013161
False			True	True	
-4.918105e-03			0.003481	8.398659e-03	

29	90	1.025295e-01	0.008535	0.010304	0.009809
False			False	True	
-1.768585e-03			-0.000494	1.274327e-03	
29	100	9.169122e-02	0.006838	0.006159	0.009763
True			True	True	
6.796914e-04			0.003604	2.924217e-03	
29	110	9.910014e-02	0.009511	0.007446	0.009636
True			True	True	
2.064948e-03			0.002190	1.250598e-04	
29	120	1.052639e-01	0.008066	0.008652	0.006965
False			False	False	
-5.854030e-04			-0.001687	-1.101757e-03	
29	130	1.088502e-01	0.007001	0.005790	0.005828
True			True	False	
1.211477e-03			0.000039	-1.172789e-03	
29	140	6.362128e-02	0.006135	0.005277	0.004805
True			False	False	
8.582951e-04			-0.000471	-1.329643e-03	
29	150	5.822491e-02	0.008037	0.002633	0.003002
True			True	False	
5.404623e-03			0.000369	-5.035787e-03	
30	10	1.339948e+00	0.747336	0.791183	0.076870
False			False	False	
-4.384700e-02			-0.714313	-6.704664e-01	
30	20	2.608839e-01	0.114449	0.047584	0.062774
True			True	False	
6.686525e-02			0.015191	-5.167463e-02	
30	30	2.034475e-01	0.038035	0.039481	0.041762
False			True	True	
-1.445754e-03			0.002281	3.727188e-03	
30	40	2.162753e-01	0.030958	0.019097	0.029965
True			True	False	
1.186078e-02			0.010868	-9.929665e-04	
30	50	1.298349e-01	0.021003	0.019043	0.019389
True			True	False	
1.960630e-03			0.000347	-1.614092e-03	
30	60	1.096579e-01	0.013912	0.015069	0.012758
False			False	False	
-1.157021e-03			-0.002310	-1.153276e-03	
30	70	9.026638e-02	0.014729	0.012736	0.011789
True			False	False	
1.993116e-03			-0.000947	-2.940228e-03	
30	80	1.034334e-01	0.011174	0.014278	0.006837
False			False	False	
-3.103510e-03			-0.007441	-4.337065e-03	
30	90	9.461454e-02	0.011948	0.007124	0.008024
True			True	False	
4.823626e-03			0.000900	-3.923581e-03	

30	100	9.679963e-02	0.005745	0.005404	0.009122
True			True	True	
3.405429e-04			0.003718	3.376987e-03	
30	110	8.167105e-02	0.005130	0.004709	0.007547
True			True	True	
4.206830e-04			0.002838	2.416966e-03	
30	120	6.756874e-02	0.004123	0.009073	0.006495
False			False	True	
-4.949092e-03			-0.002578	2.371168e-03	
30	130	6.804904e-02	0.004385	0.005064	0.005528
False			True	True	
-6.785600e-04			0.000464	1.142881e-03	
30	140	7.788816e-02	0.003665	0.003284	0.004814
True			True	True	
3.802547e-04			0.001530	1.149329e-03	
30	150	7.094604e-02	0.007487	0.004395	0.004485
True			True	False	
3.092718e-03			0.000091	-3.001824e-03	
31	10	6.160848e-01	0.699503	0.774399	0.664041
False			False	False	
-7.489608e-02			-0.110358	-3.546213e-02	
31	20	3.616196e-01	0.047726	0.034275	0.082200
True			True	True	
1.345083e-02			0.047925	3.447452e-02	
31	30	3.117688e-01	0.045690	0.029828	0.020346
True			False	False	
1.586275e-02			-0.009481	-2.534400e-02	
31	40	1.456137e-01	0.026586	0.011369	0.014873
True			True	False	
1.521663e-02			0.003504	-1.171294e-02	
31	50	1.064103e-01	0.018242	0.015476	0.011762
True			False	False	
2.766119e-03			-0.003713	-6.479501e-03	
31	60	8.611650e-02	0.013864	0.012687	0.016983
True			True	True	
1.176673e-03			0.004296	3.119249e-03	
31	70	8.408334e-02	0.007743	0.006440	0.012006
True			True	True	
1.303233e-03			0.005566	4.262714e-03	
31	80	9.333724e-02	0.009133	0.009050	0.012354
True			True	True	
8.246723e-05			0.003303	3.220951e-03	
31	90	1.286188e-01	0.008383	0.008183	0.009255
True			True	True	
1.998857e-04			0.001071	8.714799e-04	
31	100	1.191446e-01	0.013123	0.004053	0.008530
True			True	False	
9.069808e-03			0.004478	-4.592151e-03	

31	110	9.113028e-02	0.010793	0.006061	0.007400
True			True	False	
4.732121e-03			0.001339	-3.393157e-03	
31	120	9.846523e-02	0.010234	0.002979	0.005125
True			True	False	
7.255448e-03			0.002146	-5.109453e-03	
31	130	8.491586e-02	0.009457	0.003115	0.006601
True			True	False	
6.341834e-03			0.003486	-2.855863e-03	
31	140	7.393946e-02	0.008004	0.002739	0.005772
True			True	False	
5.264980e-03			0.003033	-2.232062e-03	
31	150	8.800454e-02	0.005189	0.004210	0.005039
True			True	False	
9.790332e-04			0.000830	-1.493599e-04	
32	10	1.043232e+00	0.726068	0.726068	0.105214
True			False	False	
1.110223e-16			-0.620854	-6.208536e-01	
32	20	3.827479e-01	0.063824	0.061176	0.035705
True			False	False	
2.647724e-03			-0.025472	-2.811937e-02	
32	30	1.685958e-01	0.027932	0.021945	0.023844
True			True	False	
5.986298e-03			0.001898	-4.088156e-03	
32	40	2.478030e-01	0.018970	0.017712	0.017527
True			False	False	
1.258537e-03			-0.000185	-1.443127e-03	
32	50	2.508208e-01	0.017006	0.013884	0.015830
True			True	False	
3.122121e-03			0.001946	-1.176534e-03	
32	60	2.464603e-01	0.014469	0.010691	0.016352
True			True	True	
3.778449e-03			0.005661	1.882753e-03	
32	70	2.212333e-01	0.011036	0.008396	0.013768
True			True	True	
2.639871e-03			0.005372	2.732354e-03	
32	80	1.683639e-01	0.009034	0.009443	0.013213
False			True	True	
-4.089848e-04			0.003769	4.178312e-03	
32	90	1.393394e-01	0.009441	0.009901	0.013466
False			True	True	
-4.594232e-04			0.003565	4.024766e-03	
32	100	1.127967e-01	0.008340	0.006298	0.010799
True			True	True	
2.041672e-03			0.004501	2.458873e-03	
32	110	1.011963e-01	0.008531	0.008894	0.007289
False			False	False	
-3.626188e-04			-0.001605	-1.242598e-03	

32	120	1.051177e-01	0.006405	0.004556	0.006609
True			True	True	
1.848174e-03			0.002052	2.040045e-04	
32	130	1.079910e-01	0.006504	0.004085	0.006425
True			True	False	
2.419262e-03			0.002340	-7.909289e-05	
32	140	6.927498e-02	0.005903	0.003574	0.005391
True			True	False	
2.329270e-03			0.001817	-5.126041e-04	
32	150	7.092168e-02	0.005747	0.002023	0.004555
True			True	False	
3.724367e-03			0.002532	-1.192383e-03	
33	10	1.099561e+00	0.695916	0.791520	0.806221
False			True	True	
-9.560344e-02			0.014701	1.103043e-01	
33	20	3.629355e-01	0.047588	0.080380	0.039284
False			False	False	
-3.279236e-02			-0.041096	-8.303538e-03	
33	30	2.562604e-01	0.023289	0.020002	0.036566
True			True	True	
3.286679e-03			0.016564	1.327728e-02	
33	40	1.919983e-01	0.015333	0.025465	0.022701
False			False	True	
-1.013199e-02			-0.002764	7.367771e-03	
33	50	1.817897e-01	0.019580	0.014784	0.013287
True			False	False	
4.795813e-03			-0.001497	-6.293148e-03	
33	60	1.479643e-01	0.009429	0.005724	0.018252
True			True	True	
3.705139e-03			0.012528	8.822912e-03	
33	70	1.619752e-01	0.007955	0.011047	0.009522
False			False	True	
-3.092322e-03			-0.001525	1.567406e-03	
33	80	1.407551e-01	0.008552	0.007367	0.009134
True			True	True	
1.184986e-03			0.001767	5.822586e-04	
33	90	1.026221e-01	0.006261	0.010163	0.009093
False			False	True	
-3.901892e-03			-0.001070	2.831924e-03	
33	100	8.848830e-02	0.008066	0.006791	0.009807
True			True	True	
1.274946e-03			0.003016	1.740876e-03	
33	110	9.307097e-02	0.008220	0.004205	0.007954
True			True	False	
4.015297e-03			0.003749	-2.663512e-04	
33	120	9.027925e-02	0.005277	0.005740	0.007183
False			True	True	
-4.635109e-04			0.001443	1.906452e-03	

33	130	8.087876e-02	0.004704	0.004071	0.007121
True			True	True	
6.324831e-04			0.003050	2.417679e-03	
33	140	5.229415e-02	0.005437	0.004361	0.007041
True			True	True	
1.075680e-03			0.002679	1.603806e-03	
33	150	5.346082e-02	0.004852	0.004550	0.006852
True			True	True	
3.018100e-04			0.002302	2.000197e-03	
34	10	6.409168e-01	0.632183	0.810338	0.069731
False			False	False	
-1.781554e-01			-0.740607	-5.624516e-01	
34	20	2.761474e-01	0.031285	0.044570	0.107660
False			True	True	
-1.328425e-02			0.063091	7.637520e-02	
34	30	1.881658e-01	0.037003	0.031241	0.044461
True			True	True	
5.762026e-03			0.013220	7.457584e-03	
34	40	2.178285e-01	0.040417	0.033043	0.023509
True			False	False	
7.374010e-03			-0.009535	-1.690857e-02	
34	50	2.064640e-01	0.025987	0.017274	0.022926
True			True	False	
8.713394e-03			0.005653	-3.060870e-03	
34	60	1.868410e-01	0.015514	0.004315	0.011654
True			True	False	
1.119844e-02			0.007338	-3.860131e-03	
34	70	2.018723e-01	0.013536	0.010384	0.010552
True			True	False	
3.151369e-03			0.000168	-2.983740e-03	
34	80	1.302460e-01	0.007573	0.006795	0.010853
True			True	True	
7.784381e-04			0.004059	3.280397e-03	
34	90	1.147399e-01	0.007929	0.007557	0.007701
True			True	False	
3.720407e-04			0.000143	-2.287498e-04	
34	100	6.560288e-02	0.008940	0.009046	0.008589
False			False	False	
-1.063037e-04			-0.000457	-3.504899e-04	
34	110	4.182778e-02	0.007862	0.008072	0.007712
False			False	False	
-2.101862e-04			-0.000360	-1.500874e-04	
34	120	4.858764e-02	0.007848	0.007550	0.009049
True			True	True	
2.977339e-04			0.001500	1.201883e-03	
34	130	6.070350e-02	0.007198	0.007938	0.007677
False			False	True	
-7.402985e-04			-0.000261	4.789268e-04	

34	140	8.551767e-02	0.008619	0.006618	0.006312
True			False		False
2.001844e-03			-0.000305	-2.307178e-03	
34	150	7.129427e-02	0.007100	0.002490	0.006027
True			True		False
4.610711e-03			0.003538	-1.073191e-03	
35	10	4.396766e-01	0.736909	0.046411	0.813425
True			True		True
6.904975e-01			0.767014	7.651615e-02	
35	20	3.258109e-01	0.077005	0.028927	0.039331
True			True		False
4.807711e-02			0.010403	-3.767394e-02	
35	30	2.834536e-01	0.041781	0.046196	0.055490
False			True		True
-4.415005e-03			0.009294	1.370918e-02	
35	40	1.888871e-01	0.037621	0.016856	0.014048
True			False		False
2.076530e-02			-0.002808	-2.357311e-02	
35	50	1.358809e-01	0.035449	0.014376	0.021094
True			True		False
2.107276e-02			0.006718	-1.435507e-02	
35	60	1.386550e-01	0.013674	0.009239	0.008415
True			False		False
4.434884e-03			-0.000824	-5.258858e-03	
35	70	1.061712e-01	0.009345	0.011873	0.009733
False			False		True
-2.528854e-03			-0.002141	3.881858e-04	
35	80	9.360409e-02	0.009999	0.005063	0.009031
True			True		False
4.935458e-03			0.003968	-9.670983e-04	
35	90	6.741428e-02	0.009383	0.009581	0.008806
False			False		False
-1.978762e-04			-0.000775	-5.770446e-04	
35	100	5.363513e-02	0.009191	0.007999	0.007402
True			False		False
1.191965e-03			-0.000596	-1.788204e-03	
35	110	5.817774e-02	0.010050	0.008752	0.007202
True			False		False
1.298136e-03			-0.001549	-2.847566e-03	
35	120	6.697047e-02	0.007945	0.007035	0.007392
True			True		False
9.104896e-04			0.000357	-5.534646e-04	
35	130	7.502019e-02	0.007895	0.005098	0.006469
True			True		False
2.796804e-03			0.001371	-1.426301e-03	
35	140	6.600175e-02	0.004791	0.004887	0.005822
False			True		True
-9.595957e-05			0.000935	1.030551e-03	

35	150	7.576379e-02	0.007033	0.004900	0.004269
True			False		False
2.132455e-03			-0.000632	-2.764170e-03	
36	10	1.414657e+00	0.654808	0.504442	0.689001
True			True		True
1.503652e-01			0.184558	3.419317e-02	
36	20	4.339234e-01	0.057471	0.078795	0.033474
False			False		False
-2.132387e-02			-0.045321	-2.399730e-02	
36	30	2.077498e-01	0.056304	0.035123	0.022031
True			False		False
2.118114e-02			-0.013092	-3.427303e-02	
36	40	2.189675e-01	0.030533	0.015099	0.022139
True			True		False
1.543351e-02			0.007039	-8.394195e-03	
36	50	1.643609e-01	0.016944	0.011789	0.027782
True			True		True
5.155176e-03			0.015993	1.083790e-02	
36	60	1.701612e-01	0.017421	0.017896	0.012857
False			False		False
-4.746283e-04			-0.005038	-4.563789e-03	
36	70	1.775017e-01	0.013368	0.009782	0.010620
True			True		False
3.585277e-03			0.000837	-2.747951e-03	
36	80	1.722787e-01	0.009610	0.005802	0.008557
True			True		False
3.808405e-03			0.002755	-1.052945e-03	
36	90	1.329279e-01	0.006453	0.005701	0.009368
True			True		True
7.523062e-04			0.003667	2.914641e-03	
36	100	9.693845e-02	0.005984	0.005653	0.007015
True			True		True
3.313486e-04			0.001362	1.030511e-03	
36	110	4.783551e-02	0.005550	0.005239	0.006548
True			True		True
3.106552e-04			0.001309	9.981355e-04	
36	120	5.683110e-02	0.004918	0.005607	0.007410
False			True		True
-6.896275e-04			0.001803	2.492417e-03	
36	130	6.182677e-02	0.004773	0.005328	0.006695
False			True		True
-5.546513e-04			0.001367	1.921846e-03	
36	140	5.527682e-02	0.005502	0.005362	0.006333
True			True		True
1.404248e-04			0.000971	8.304948e-04	
36	150	5.546819e-02	0.003396	0.004558	0.004723
False			True		True
-1.162629e-03			0.000165	1.327664e-03	

37	10	4.835140e-01	0.558237	0.809900	0.821371
False			True	True	
-2.516629e-01			0.011472	2.631347e-01	
37	20	2.742920e-01	0.033327	0.040449	0.080091
False			True	True	
-7.122663e-03			0.039642	4.676424e-02	
37	30	2.635778e-01	0.024601	0.033118	0.034358
False			True	True	
-8.517499e-03			0.001239	9.756553e-03	
37	40	1.962596e-01	0.019964	0.024438	0.023860
False			False	True	
-4.474141e-03			-0.000578	3.895976e-03	
37	50	1.434879e-01	0.009794	0.016381	0.018382
False			True	True	
-6.587003e-03			0.002001	8.587563e-03	
37	60	1.508768e-01	0.011929	0.011685	0.009070
True			False	False	
2.439117e-04			-0.002615	-2.858685e-03	
37	70	1.318588e-01	0.016578	0.012441	0.008985
True			False	False	
4.136349e-03			-0.003456	-7.592746e-03	
37	80	8.640129e-02	0.018535	0.013248	0.011429
True			False	False	
5.287177e-03			-0.001819	-7.105685e-03	
37	90	6.681215e-02	0.013247	0.010374	0.008944
True			False	False	
2.873095e-03			-0.001430	-4.303055e-03	
37	100	6.473924e-02	0.010444	0.008713	0.007406
True			False	False	
1.730535e-03			-0.001308	-3.038266e-03	
37	110	7.554952e-02	0.009800	0.007175	0.007273
True			True	False	
2.625524e-03			0.000098	-2.527048e-03	
37	120	1.071665e-01	0.007907	0.006908	0.006083
True			False	False	
9.982435e-04			-0.000825	-1.823493e-03	
37	130	1.037711e-01	0.003429	0.003996	0.007087
False			True	True	
-5.672705e-04			0.003090	3.657589e-03	
37	140	8.510571e-02	0.005191	0.006440	0.006704
False			True	True	
-1.248223e-03			0.000264	1.512382e-03	
37	150	7.834673e-02	0.006710	0.006184	0.006153
True			False	False	
5.255877e-04			-0.000031	-5.569976e-04	
38	10	9.461336e-01	0.105140	0.515082	0.087108
False			False	False	
-4.099417e-01			-0.427974	-1.803250e-02	

38	20	4.330950e-01	0.047185	0.042408	0.081086
True			True	True	
4.777139e-03			0.038678	3.390045e-02	
38	30	3.296969e-01	0.029545	0.025809	0.022268
True			False	False	
3.736228e-03			-0.003541	-7.276944e-03	
38	40	1.822136e-01	0.025747	0.016534	0.020303
True			True	False	
9.212282e-03			0.003768	-5.443856e-03	
38	50	1.540819e-01	0.011811	0.007805	0.021212
True			True	True	
4.006078e-03			0.013406	9.400178e-03	
38	60	1.159685e-01	0.011713	0.013828	0.010781
False			False	False	
-2.115447e-03			-0.003047	-9.318250e-04	
38	70	9.157067e-02	0.009387	0.007027	0.006851
True			False	False	
2.359932e-03			-0.000176	-2.535728e-03	
38	80	9.512271e-02	0.010269	0.011891	0.007660
False			False	False	
-1.622245e-03			-0.004231	-2.608612e-03	
38	90	9.518403e-02	0.012678	0.007138	0.008575
True			True	False	
5.540809e-03			0.001437	-4.103627e-03	
38	100	9.519824e-02	0.008311	0.005989	0.007858
True			True	False	
2.321922e-03			0.001869	-4.524366e-04	
38	110	1.124803e-01	0.005653	0.005716	0.005666
False			False	True	
-6.260787e-05			-0.000049	1.346566e-05	
38	120	1.060997e-01	0.005471	0.007737	0.006692
False			False	True	
-2.266602e-03			-0.001045	1.221371e-03	
38	130	9.088930e-02	0.006473	0.004566	0.007155
True			True	True	
1.907850e-03			0.002589	6.816306e-04	
38	140	8.441160e-02	0.004292	0.003458	0.006413
True			True	True	
8.343691e-04			0.002955	2.120612e-03	
38	150	6.677499e-02	0.002803	0.004167	0.006044
False			True	True	
-1.364815e-03			0.001877	3.241798e-03	
39	10	7.011778e+06	0.526795	0.791582	0.489647
False			False	False	
-2.647873e-01			-0.301935	-3.714722e-02	
39	20	3.410739e-01	0.045557	0.025446	0.068720
True			True	True	
2.011039e-02			0.043274	2.316363e-02	

39	30	2.837704e-01	0.020323	0.030184	0.026100
False			False		True
-9.861816e-03			-0.004084	5.777570e-03	
39	40	1.628845e-01	0.019752	0.021244	0.027714
False			True		True
-1.491786e-03			0.006470	7.961386e-03	
39	50	2.046779e-01	0.034136	0.012646	0.022940
True			True		False
2.148998e-02			0.010294	-1.119570e-02	
39	60	8.649475e-02	0.014167	0.015175	0.013680
False			False		False
-1.007709e-03			-0.001495	-4.871212e-04	
39	70	6.464374e-02	0.010472	0.009114	0.015027
True			True		True
1.358011e-03			0.005913	4.555433e-03	
39	80	5.012401e-02	0.005375	0.010148	0.014190
False			True		True
-4.772403e-03			0.004042	8.814415e-03	
39	90	4.440656e-02	0.008576	0.008798	0.013110
False			True		True
-2.222512e-04			0.004312	4.534420e-03	
39	100	4.274634e-02	0.009654	0.007506	0.011779
True			True		True
2.147962e-03			0.004273	2.124989e-03	
39	110	5.065900e-02	0.007646	0.007300	0.006941
True			False		False
3.459649e-04			-0.000359	-7.047369e-04	
39	120	5.388702e-02	0.006256	0.006991	0.007231
False			True		True
-7.344366e-04			0.000240	9.743870e-04	
39	130	5.161889e-02	0.006095	0.006058	0.006358
True			True		True
3.772829e-05			0.000301	2.628009e-04	
39	140	4.733112e-02	0.004325	0.004443	0.004973
False			True		True
-1.182630e-04			0.000530	6.479901e-04	
39	150	4.690699e-02	0.002135	0.004954	0.004228
False			False		True
-2.818586e-03			-0.000726	2.092998e-03	
40	10	1.075167e+00	0.521464	0.068937	0.772787
True			True		True
4.525268e-01			0.703850	2.513229e-01	
40	20	6.703467e-01	0.036222	0.049710	0.053561
False			True		True
-1.348818e-02			0.003851	1.733903e-02	
40	30	2.585133e-01	0.025327	0.026503	0.023808
False			False		False
-1.176964e-03			-0.002696	-1.518594e-03	

40	40	2.632796e-01	0.024382	0.021486	0.010501
True			False		False
2.896469e-03			-0.010985	-1.388121e-02	
40	50	2.008882e-01	0.012977	0.019918	0.010712
False			False		False
-6.941401e-03			-0.009206	-2.264117e-03	
40	60	1.685649e-01	0.008778	0.009995	0.012059
False			True		True
-1.217583e-03			0.002064	3.281483e-03	
40	70	1.710687e-01	0.010720	0.007068	0.009175
True			True		False
3.652543e-03			0.002107	-1.545514e-03	
40	80	1.492562e-01	0.010387	0.006223	0.005348
True			False		False
4.163486e-03			-0.000875	-5.038823e-03	
40	90	1.350653e-01	0.011330	0.007941	0.006425
True			False		False
3.388693e-03			-0.001516	-4.904596e-03	
40	100	1.363467e-01	0.008179	0.007486	0.007163
True			False		False
6.934673e-04			-0.000323	-1.016101e-03	
40	110	1.264832e-01	0.006402	0.007349	0.007183
False			False		True
-9.470496e-04			-0.000166	7.811259e-04	
40	120	1.067744e-01	0.005253	0.004248	0.007110
True			True		True
1.004955e-03			0.002861	1.856499e-03	
40	130	9.059909e-02	0.006411	0.005100	0.006430
True			True		True
1.310960e-03			0.001330	1.935694e-05	
40	140	6.911812e-02	0.006801	0.003971	0.005386
True			True		False
2.830568e-03			0.001415	-1.415145e-03	
40	150	7.381667e-02	0.004806	0.006076	0.004805
False			False		False
-1.270624e-03			-0.001271	-5.081273e-07	
41	10	7.980126e-01	0.086494	0.069535	0.814171
True			True		True
1.695943e-02			0.744636	7.276770e-01	
41	20	3.492994e-01	0.040866	0.034934	0.709178
True			True		True
5.932114e-03			0.674244	6.683124e-01	
41	30	3.444982e-01	0.055456	0.045738	0.022999
True			False		False
9.717736e-03			-0.022739	-3.245625e-02	
41	40	1.413946e-01	0.036522	0.028498	0.021179
True			False		False
8.024199e-03			-0.007319	-1.534317e-02	

41	50	1.810138e-01	0.023839	0.015390	0.022782
True			True	False	
8.449709e-03			0.007393	-1.057122e-03	
41	60	1.971385e-01	0.012821	0.012110	0.016694
True			True	True	
7.106484e-04			0.004583	3.872722e-03	
41	70	1.580656e-01	0.009607	0.007584	0.012742
True			True	True	
2.023651e-03			0.005158	3.134598e-03	
41	80	1.382061e-01	0.007380	0.006491	0.009778
True			True	True	
8.891760e-04			0.003287	2.397482e-03	
41	90	1.158350e-01	0.007661	0.009783	0.010209
False			True	True	
-2.122552e-03			0.000426	2.548312e-03	
41	100	9.321317e-02	0.009763	0.006014	0.009270
True			True	False	
3.748394e-03			0.003256	-4.921151e-04	
41	110	6.365288e-02	0.006989	0.004821	0.008556
True			True	True	
2.168458e-03			0.003736	1.567251e-03	
41	120	6.375679e-02	0.006343	0.005278	0.008314
True			True	True	
1.064501e-03			0.003036	1.971105e-03	
41	130	7.905284e-02	0.006316	0.005869	0.007603
True			True	True	
4.469403e-04			0.001734	1.287036e-03	
41	140	6.167792e-02	0.005170	0.003833	0.006812
True			True	True	
1.337645e-03			0.002979	1.641722e-03	
41	150	5.513581e-02	0.004197	0.004747	0.006429
False			True	True	
-5.502243e-04			0.001682	2.232104e-03	
42	10	5.349879e-01	0.786076	0.100076	0.787809
True			True	True	
6.860001e-01			0.687733	1.733258e-03	
42	20	3.542155e-01	0.041180	0.043623	0.651373
False			True	True	
-2.443062e-03			0.607750	6.101931e-01	
42	30	1.197387e-01	0.023187	0.020315	0.028491
True			True	True	
2.872084e-03			0.008175	5.303260e-03	
42	40	1.697367e-01	0.017516	0.023545	0.024696
False			True	True	
-6.029258e-03			0.001151	7.180651e-03	
42	50	1.440292e-01	0.013826	0.021047	0.018932
False			False	True	
-7.221233e-03			-0.002115	5.106206e-03	

42	60	1.754884e-01	0.008346	0.008745	0.014835
False			True	True	
-3.990197e-04			0.006090	6.489020e-03	
42	70	1.516651e-01	0.011084	0.011799	0.013539
False			True	True	
-7.151920e-04			0.001740	2.454944e-03	
42	80	1.193150e-01	0.012391	0.010940	0.011451
True			True	False	
1.451230e-03			0.000510	-9.409728e-04	
42	90	1.113029e-01	0.009766	0.010752	0.012678
False			True	True	
-9.852621e-04			0.001926	2.911449e-03	
42	100	8.714739e-02	0.010974	0.004462	0.013053
True			True	True	
6.512369e-03			0.008591	2.078636e-03	
42	110	7.661287e-02	0.008642	0.006801	0.013248
True			True	True	
1.840624e-03			0.006447	4.605914e-03	
42	120	5.647849e-02	0.008052	0.005552	0.010975
True			True	True	
2.500036e-03			0.005423	2.922876e-03	
42	130	4.515906e-02	0.009253	0.006848	0.011534
True			True	True	
2.405815e-03			0.004686	2.280432e-03	
42	140	7.852624e-02	0.008479	0.004997	0.008682
True			True	True	
3.482666e-03			0.003685	2.024603e-04	
42	150	8.282318e-02	0.005525	0.003981	0.008705
True			True	True	
1.543502e-03			0.004724	3.180561e-03	
43	10	7.309919e-01	0.727329	0.775491	0.739735
False			False	True	
-4.816198e-02			-0.035755	1.240648e-02	
43	20	4.341430e-01	0.038156	0.676006	0.020956
False			False	False	
-6.378497e-01			-0.655049	-1.719960e-02	
43	30	2.079319e-01	0.043972	0.025631	0.021485
True			False	False	
1.834168e-02			-0.004145	-2.248708e-02	
43	40	2.362832e-01	0.034948	0.013576	0.017789
True			True	False	
2.137169e-02			0.004213	-1.715846e-02	
43	50	1.507086e-01	0.023380	0.014733	0.021212
True			True	False	
8.647381e-03			0.006479	-2.168378e-03	
43	60	1.585529e-01	0.014102	0.009146	0.017551
True			True	True	
4.955911e-03			0.008405	3.448975e-03	

43	70	1.337825e-01	0.014685	0.009033	0.015277
True			True	True	
5.652037e-03			0.006244	5.919840e-04	
43	80	1.287486e-01	0.014599	0.012560	0.013226
True			True	False	
2.039101e-03			0.000666	-1.372910e-03	
43	90	1.012347e-01	0.014205	0.010948	0.012163
True			True	False	
3.257303e-03			0.001215	-2.042750e-03	
43	100	7.185575e-02	0.013080	0.005278	0.012585
True			True	False	
7.801797e-03			0.007307	-4.944157e-04	
43	110	5.181460e-02	0.014499	0.007045	0.009724
True			True	False	
7.453416e-03			0.002679	-4.774256e-03	
43	120	5.984578e-02	0.010537	0.003629	0.008197
True			True	False	
6.907636e-03			0.004568	-2.339823e-03	
43	130	5.974856e-02	0.009066	0.006177	0.008546
True			True	False	
2.889200e-03			0.002369	-5.204730e-04	
43	140	7.356726e-02	0.007751	0.003179	0.007164
True			True	False	
4.571987e-03			0.003984	-5.876789e-04	
43	150	5.327531e-02	0.007468	0.003712	0.005943
True			True	False	
3.756592e-03			0.002231	-1.525160e-03	
44	10	7.546040e-01	0.742227	0.548709	0.775033
True			True	True	
1.935179e-01			0.226324	3.280596e-02	
44	20	2.341309e-01	0.700608	0.034577	0.046740
True			True	False	
6.660311e-01			0.012164	-6.538674e-01	
44	30	1.472041e-01	0.041391	0.024844	0.022270
True			False	False	
1.654770e-02			-0.002574	-1.912148e-02	
44	40	2.929831e-01	0.025913	0.039482	0.029224
False			False	True	
-1.356955e-02			-0.010258	3.311617e-03	
44	50	2.187787e-01	0.026552	0.013878	0.017196
True			True	False	
1.267337e-02			0.003318	-9.355598e-03	
44	60	1.509165e-01	0.011323	0.010024	0.009258
True			False	False	
1.299532e-03			-0.000766	-2.065142e-03	
44	70	1.007508e-01	0.015094	0.014522	0.008290
True			False	False	
5.717106e-04			-0.006232	-6.804183e-03	

44	80	1.087249e-01	0.013306	0.009141	0.006864
True			False		False
4.164657e-03			-0.002278	-6.442394e-03	
44	90	1.058828e-01	0.010760	0.004681	0.006628
True			True		False
6.079030e-03			0.001946	-4.132559e-03	
44	100	9.956598e-02	0.012719	0.004740	0.006195
True			True		False
7.979483e-03			0.001455	-6.524466e-03	
44	110	9.125272e-02	0.005050	0.005487	0.006791
False			True		True
-4.370828e-04			0.001304	1.741178e-03	
44	120	6.718147e-02	0.005369	0.004593	0.006720
True			True		True
7.764760e-04			0.002127	1.350755e-03	
44	130	6.192102e-02	0.005443	0.003143	0.006250
True			True		True
2.300722e-03			0.003107	8.066514e-04	
44	140	7.140767e-02	0.002090	0.003735	0.007433
False			True		True
-1.644965e-03			0.003698	5.343141e-03	
44	150	7.863257e-02	0.007921	0.004364	0.006438
True			True		False
3.557716e-03			0.002075	-1.482917e-03	
45	10	5.051973e-01	0.731119	0.297248	0.084862
True			False		False
4.338710e-01			-0.212387	-6.462576e-01	
45	20	2.898617e-01	0.041933	0.038156	0.106299
True			True		True
3.777726e-03			0.068143	6.436537e-02	
45	30	2.931705e-01	0.031572	0.024947	0.031928
True			True		True
6.624584e-03			0.006981	3.559930e-04	
45	40	2.074835e-01	0.027423	0.030499	0.029435
False			False		True
-3.075529e-03			-0.001064	2.011803e-03	
45	50	1.265342e-01	0.016464	0.012507	0.024619
True			True		True
3.956933e-03			0.012111	8.154148e-03	
45	60	1.008858e-01	0.016966	0.013552	0.012974
True			False		False
3.413227e-03			-0.000578	-3.991372e-03	
45	70	8.445935e-02	0.013472	0.004457	0.012340
True			True		False
9.015462e-03			0.007883	-1.132252e-03	
45	80	5.526071e-02	0.011436	0.006459	0.009368
True			True		False
4.977040e-03			0.002908	-2.068861e-03	

45	90	5.102912e-02	0.006880	0.006796	0.009060
True			True	True	
8.414758e-05			0.002264	2.179815e-03	
45	100	9.133120e-02	0.009415	0.006825	0.007809
True			True	False	
2.590063e-03			0.000984	-1.606166e-03	
45	110	9.243665e-02	0.006834	0.004119	0.006174
True			True	False	
2.715231e-03			0.002055	-6.607060e-04	
45	120	6.268644e-02	0.005019	0.001947	0.005643
True			True	True	
3.072451e-03			0.003696	6.238432e-04	
45	130	5.294894e-02	0.006378	0.003662	0.005809
True			True	False	
2.715932e-03			0.002148	-5.683572e-04	
45	140	3.584037e-02	0.002682	0.002768	0.006364
False			True	True	
-8.643592e-05			0.003596	3.681989e-03	
45	150	3.413746e-02	0.006158	0.003594	0.005597
True			True	False	
2.563732e-03			0.002003	-5.608108e-04	
46	10	7.447560e-01	0.541932	0.774077	0.810922
False			True	True	
-2.321445e-01			0.036845	2.689896e-01	
46	20	5.642989e-01	0.022240	0.052689	0.071027
False			True	True	
-3.044905e-02			0.018337	4.878629e-02	
46	30	3.348064e-01	0.021902	0.014042	0.035434
True			True	True	
7.860596e-03			0.021392	1.353167e-02	
46	40	2.484350e-01	0.038855	0.024495	0.033866
True			True	False	
1.435953e-02			0.009371	-4.988993e-03	
46	50	2.464161e-01	0.039265	0.010549	0.027335
True			True	False	
2.871514e-02			0.016785	-1.192966e-02	
46	60	2.069327e-01	0.010697	0.011088	0.014624
False			True	True	
-3.909852e-04			0.003536	3.926517e-03	
46	70	1.704269e-01	0.017733	0.010548	0.011628
True			True	False	
7.185059e-03			0.001079	-6.105729e-03	
46	80	1.437160e-01	0.014246	0.014455	0.006309
False			False	False	
-2.089213e-04			-0.008147	-7.937757e-03	
46	90	1.309961e-01	0.013347	0.011433	0.005240
True			False	False	
1.913783e-03			-0.006193	-8.106783e-03	

46	100	1.172121e-01	0.008726	0.008910	0.003618
False			False	False	
-1.845302e-04			-0.005292	-5.107970e-03	
46	110	9.802305e-02	0.009689	0.005182	0.003703
True			False	False	
4.507308e-03			-0.001479	-5.986026e-03	
46	120	9.173154e-02	0.010785	0.004980	0.003585
True			False	False	
5.805094e-03			-0.001395	-7.200147e-03	
46	130	8.709751e-02	0.009614	0.006488	0.003241
True			False	False	
3.125626e-03			-0.003247	-6.373083e-03	
46	140	6.787087e-02	0.007218	0.003231	0.002880
True			False	False	
3.986889e-03			-0.000351	-4.337703e-03	
46	150	4.312594e-02	0.004950	0.004354	0.003182
True			False	False	
5.954499e-04			-0.001172	-1.767642e-03	
47	10	7.056846e-01	0.651582	0.696079	0.693089
False			False	True	
-4.449745e-02			-0.002990	4.150779e-02	
47	20	6.220921e-01	0.105373	0.060535	0.055430
True			False	False	
4.483848e-02			-0.005105	-4.994328e-02	
47	30	3.288609e-01	0.024978	0.032210	0.028629
False			False	True	
-7.232257e-03			-0.003581	3.650908e-03	
47	40	1.888458e-01	0.018997	0.018062	0.019261
True			True	True	
9.356880e-04			0.001200	2.640652e-04	
47	50	1.228822e-01	0.020878	0.019321	0.024514
True			True	True	
1.556668e-03			0.005192	3.635471e-03	
47	60	1.488832e-01	0.016027	0.012979	0.013564
True			True	False	
3.047149e-03			0.000585	-2.462621e-03	
47	70	1.792881e-01	0.014810	0.012578	0.013514
True			True	False	
2.232199e-03			0.000936	-1.296028e-03	
47	80	1.365095e-01	0.013684	0.007938	0.013834
True			True	True	
5.746691e-03			0.005897	1.498428e-04	
47	90	1.340818e-01	0.012521	0.008579	0.011825
True			True	False	
3.941145e-03			0.003246	-6.951260e-04	
47	100	1.110583e-01	0.010804	0.008542	0.009433
True			True	False	
2.261182e-03			0.000891	-1.370193e-03	

47	110	1.031316e-01	0.006224	0.007287	0.008436
False			True	True	
-1.062926e-03			0.001149	2.211949e-03	
47	120	6.116201e-02	0.005754	0.010072	0.007937
False			False	True	
-4.317591e-03			-0.002135	2.182920e-03	
47	130	5.826831e-02	0.006439	0.008351	0.007038
False			False	True	
-1.911679e-03			-0.001313	5.988714e-04	
47	140	4.475016e-02	0.007079	0.007204	0.007352
False			True	True	
-1.246130e-04			0.000148	2.730569e-04	
47	150	4.137668e-02	0.008558	0.005934	0.006817
True			True	False	
2.624552e-03			0.000884	-1.740795e-03	
48	10	1.765854e+00	0.103877	0.101173	0.544446
True			True	True	
2.703602e-03			0.443273	4.405698e-01	
48	20	4.963255e-01	0.070259	0.046190	0.091172
True			True	True	
2.406942e-02			0.044982	2.091256e-02	
48	30	2.823783e-01	0.026442	0.027697	0.029035
False			True	True	
-1.254877e-03			0.001338	2.592942e-03	
48	40	2.241937e-01	0.020561	0.020999	0.018641
False			False	False	
-4.383784e-04			-0.002359	-1.920317e-03	
48	50	1.721113e-01	0.013830	0.011739	0.011180
True			False	False	
2.090749e-03			-0.000559	-2.649653e-03	
48	60	1.440155e-01	0.013091	0.021272	0.012664
False			False	False	
-8.180572e-03			-0.008608	-4.270076e-04	
48	70	1.220762e-01	0.014393	0.016857	0.011233
False			False	False	
-2.463828e-03			-0.005624	-3.160358e-03	
48	80	1.210576e-01	0.018264	0.006424	0.009961
True			True	False	
1.184008e-02			0.003538	-8.302502e-03	
48	90	1.172077e-01	0.015655	0.013165	0.009952
True			False	False	
2.489873e-03			-0.003213	-5.703222e-03	
48	100	1.283258e-01	0.010849	0.008968	0.009702
True			True	False	
1.880881e-03			0.000734	-1.146536e-03	
48	110	1.204454e-01	0.013161	0.007220	0.009741
True			True	False	
5.940905e-03			0.002521	-3.419700e-03	

48	120	1.141279e-01	0.011836	0.005994	0.007092
True			True	False	
5.841964e-03			0.001098	-4.744181e-03	
48	130	9.564889e-02	0.008111	0.003701	0.007334
True			True	False	
4.410468e-03			0.003634	-7.768865e-04	
48	140	7.580182e-02	0.007527	0.003653	0.007506
True			True	False	
3.873723e-03			0.003853	-2.038848e-05	
48	150	7.125663e-02	0.007610	0.004603	0.007717
True			True	True	
3.007205e-03			0.003114	1.072496e-04	
49	10	6.207809e-01	0.588523	0.588523	0.677480
False			True	True	
-3.330669e-16			0.088957	8.895695e-02	
49	20	2.018487e-01	0.103312	0.088343	0.035685
True			False	False	
1.496979e-02			-0.052657	-6.762721e-02	
49	30	2.545506e-01	0.031935	0.036549	0.019844
False			False	False	
-4.614040e-03			-0.016705	-1.209093e-02	
49	40	2.590985e-01	0.028208	0.023714	0.019707
True			False	False	
4.494090e-03			-0.004007	-8.500708e-03	
49	50	2.013310e-01	0.024388	0.021999	0.011399
True			False	False	
2.388978e-03			-0.010600	-1.298929e-02	
49	60	1.546595e-01	0.011765	0.017596	0.015266
False			False	True	
-5.831201e-03			-0.002330	3.500990e-03	
49	70	1.161471e-01	0.018781	0.013020	0.014161
True			True	False	
5.760786e-03			0.001141	-4.619416e-03	
49	80	8.541697e-02	0.013875	0.008173	0.009844
True			True	False	
5.701364e-03			0.001670	-4.031317e-03	
49	90	8.029475e-02	0.011239	0.012829	0.011473
False			False	True	
-1.589838e-03			-0.001356	2.343012e-04	
49	100	7.011092e-02	0.009862	0.006654	0.010750
True			True	True	
3.207620e-03			0.004095	8.876605e-04	
49	110	7.487783e-02	0.011275	0.008626	0.010707
True			True	False	
2.648736e-03			0.002081	-5.677932e-04	
49	120	6.833752e-02	0.007898	0.005860	0.009476
True			True	True	
2.038100e-03			0.003616	1.577494e-03	

49	130	7.343500e-02	0.007822	0.003975	0.009487
True			True		True
3.846520e-03			0.005511	1.664668e-03	
49	140	7.855663e-02	0.007270	0.006208	0.009647
True			True		True
1.061855e-03			0.003438	2.376537e-03	
49	150	6.444093e-02	0.006299	0.005855	0.007271
True			True		True
4.444003e-04			0.001417	9.722273e-04	

=== Paired Wilcoxon significance by budget (50 shared seeds per budget) ===

budget	p_full_vs_no2c	full_better_no2c	p_full_vs_wo_bern	full_better_wo_bern
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10	0.3137	False	0.6322
False			
20	0.7304	False	0.8859
False			
30	0.3572	False	0.4964
False			
40	0.5656	False	0.9314
False			
50	0.1781	False	0.1750
False			
60	0.2115	False	0.8333
False			
70	0.0018	True	0.3572
False			
80	0.0078	True	0.4097
False			
90	0.0429	True	0.3777
False			
100	0.0255	True	0.0109
True			
110	0.0000	True	0.0005
True			
120	0.0020	True	0.0003
True			
130	0.0015	True	0.0000
True			
140	0.0002	True	0.0000
True			
150	0.0001	True	0.0000
True			

lev_full significantly better than lev_no_2c at 9/15 budgets; than
lev_wo_bernoulli at 6/15 budgets (p < 0.05, paired).

=== Overall paired Wilcoxon summary (positive margin => left model better) ===

	comparison	median_margin	p_value	left_better
	lev_full vs lev_no_2c	0.00111	1.76e-09	True
	lev_full vs lev_wo_bernoulli	0.00121	1.11e-05	True
	lev_no_2c vs lev_wo_bernoulli	-0.00035	8.22e-02	False

[8]: table

```
[8]:      run  budget  err_ks  err_lev_no_2c  err_lev_full  err_lev_wo_bernoulli  \
0      0      10  0.938719      0.701070      0.847305      0.791678
1      0      20  0.413724      0.038156      0.813323      0.093654
2      0      30  0.184986      0.045829      0.039031      0.041928
3      0      40  0.172226      0.022148      0.021287      0.016682
4      0      50  0.136241      0.019493      0.023753      0.015707
..    ...    ...
745    49     110  0.074878      0.011275      0.008626      0.010707
746    49     120  0.068338      0.007898      0.005860      0.009476
747    49     130  0.073435      0.007822      0.003975      0.009487
748    49     140  0.078557      0.007270      0.006208      0.009647
749    49     150  0.064441      0.006299      0.005855      0.007271
```

	lev_full_wins_vs_no_2c	lev_full_wins_vs_wo_bernoulli	\
0	False	False	
1	False	False	
2	True	True	
3	True	False	
4	False	False	
..	...	...	
745	True	True	
746	True	True	
747	True	True	
748	True	True	
749	True	True	

	no_2c_wins_vs_wo_bernoulli	margin_full_vs_no2c	margin_full_vs_wo_bern	\
0	True	-0.146235	-0.055626	
1	True	-0.775167	-0.719669	
2	False	0.006798	0.002897	
3	False	0.000861	-0.004605	
4	False	-0.004260	-0.008046	
..	...	...	...	
745	False	0.002649	0.002081	
746	True	0.002038	0.003616	
747	True	0.003847	0.005511	
748	True	0.001062	0.003438	
749	True	0.000444	0.001417	

margin_no2c_vs_wo_bern

```

0          0.090608
1          0.055498
2         -0.003901
3         -0.005465
4         -0.003786
..          ...
745        -0.000568
746         0.001577
747         0.001665
748         0.002377
749         0.000972

```

[750 rows x 12 columns]

```
[9]: summary_by_budget
```

```

[9]:   budget  instances  lev_full_win_count  lev_full_vs_wo_bern_win_count  \
0         10         50          22          20
1         20         50          26          27
2         30         50          27          26
3         40         50          26          25
4         50         50          30          31
5         60         50          29          23
6         70         50          35          30
7         80         50          33          32
8         90         50          30          28
9        100         50          32          33
10       110         50          34          34
11       120         50          33          33
12       130         50          34          39
13       140         50          34          38
14       150         50          38          36

```

```

   no_2c_win_count  median_err_lev_full  median_err_lev_no_2c  \
0                28          0.702597          0.641882
1                27          0.051634          0.047657
2                21          0.030713          0.030558
3                22          0.021386          0.022517
4                19          0.015743          0.018711
5                23          0.012553          0.013394
6                21          0.010398          0.012166
7                18          0.008844          0.010874
8                22          0.008905          0.009699
9                22          0.007377          0.008782
10               23          0.006780          0.008536
11               28          0.005973          0.006997
12               30          0.005236          0.006549

```

13	27	0.004522	0.006579
14	24	0.004518	0.005663

	median_err_lev_wo_bernoulli	p_full_vs_no2c	full_better_no2c \
0	0.691045	0.3137	False
1	0.052910	0.7304	False
2	0.028327	0.3572	False
3	0.021159	0.5656	False
4	0.017346	0.1781	False
5	0.013079	0.2115	False
6	0.011589	0.0018	True
7	0.009573	0.0078	True
8	0.009171	0.0429	True
9	0.009153	0.0255	True
10	0.007950	0.0000	True
11	0.007101	0.0020	True
12	0.006688	0.0015	True
13	0.006322	0.0002	True
14	0.005985	0.0001	True

	p_full_vs_wo_bern	full_better_wo_bern
0	0.6322	False
1	0.8859	False
2	0.4964	False
3	0.9314	False
4	0.1750	False
5	0.8333	False
6	0.3572	False
7	0.4097	False
8	0.3777	False
9	0.0109	True
10	0.0005	True
11	0.0003	True
12	0.0000	True
13	0.0000	True
14	0.0000	True

```
[10]: overall
```

```
[10]:
      comparison  median_margin  p_value  left_better
0  lev_full vs lev_no_2c      0.00111  1.76e-09      True
1  lev_full vs lev_wo_bernoulli  0.00121  1.11e-05      True
2  lev_no_2c vs lev_wo_bernoulli -0.00035  8.22e-02     False
```

0.0.1 Empirical Evaluation and Discussion

This benchmark isolates the effect of the $2c$ Bernoulli thresholding in LeverageSHAP by comparing three estimators on a fixed California ($n=8$) mean-substitution game:

- `lev_full` — the full LeverageSHAP (Algorithm 1) with $2c$ -based thresholding;
- `lev_no_2c` — leverage-score weights but a flat, deterministic per-size budget allocation (no $2c$ threshold);
- `lev_wo_bernoulli` — leverage-score weights sampled via the generic `CoalitionSampler` (no Algorithm 2).

On non-independence. The design uses $50 \text{ seeds} \times 15 \text{ budgets}$. The same 50 seeds recur across budgets and are shared by all methods, so the 750 (seed, budget) cells are **not** independent. We therefore do not read a raw $463/750$ “win count” as a significance statistic. Instead we run a **paired Wilcoxon signed-rank test** per budget over the 50 shared seeds, and an overall paired test pooling the per-cell differences. Raw win counts are retained only as descriptive context.

1. `lev_no_2c` vs. `lev_wo_bernoulli`: empirically similar The two non-thresholded variants track each other closely (raw win rate $\approx 47\%$, and the per-budget paired test rarely finds a consistent winner). This is expected: both sample proportional to leverage scores ($P(S) \propto 1/\binom{n}{s}$) and apply importance-sampling reweighting that cancels the $\binom{n}{s}$ term, leaving weights $\propto 1/(s(n-s))$. They are **not identical** — `lev_no_2c` uses a deterministic flat allocation while `lev_wo_bernoulli` samples stochastically — so we describe them as *empirically similar on this game* rather than as the same estimator.

2. Effect of the $2c$ threshold (`lev_full`) Whether `lev_full` improves on the non-thresholded variants is **budget-dependent** and should be read off the paired-test table above, not the raw win counts:

- At the lowest budget ($m = 10$) `lev_full` is *not* ahead: the solved threshold $2c \approx 1.14$ is below $\binom{8}{1} = 8$, so no size layer is enumerated deterministically and there is no advantage to gain.
- As m grows, $2c$ eventually exceeds $\binom{8}{1}$ and $\binom{8}{2}$, so the smallest and largest layers are taken exhaustively. Where the paired test is significant, `lev_full`’s edge comes from removing sampling variance on those high-leverage extreme-size layers.

Report the count of budgets at which the paired test is significant (printed by the analysis cell) rather than any single pooled percentage; margins at several budgets are on the order of 10^{-3} – 10^{-4} and are effectively ties.

3. Caveat This is a **single fixed instance** on one dataset (California, $n=8$). It isolates the mechanism of the $2c$ threshold but does not establish a general ranking; broader claims would require multiple instances/datasets and the paper’s interventional game.